Simulation-Test Integrated Global Optimization

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Abstract

Bayesian Optimization (BO) constitutes a canonical gradient-free global optimization paradigm, which has achieved great success in many engineering areas such as manufacturing process parameter optimization and metamaterial design. In existing theoretical developments and applications of BO, the labels of training data are generated through either physical tests or computational models. In practice, physical tests are extremely expensive to implement and suffer from measurement errors, while computational models are subjected to extensive uncertainties resulted from, e.g., the unknownness of model parameters. A new global optimization paradigm, equipped with efficient Bayesian numerical algorithms, is proposed through integrating high-cost physical tests and expensive-to-evaluate models with parameter uncertainties. The proposed framework, named Simulation-Test Integrated Global Optimization (STIGO), is fundamentally composed of two key components, i.e., Bayesian Optimal Experimental Design (BOED) and Bayesian Model Updating (BMU). For BOED, a new utility function is elaborately designed to identify, under uncertainties of both model parameters and measurements, the experimental design that is most informative to global optimum, and a Bayesian algorithm combining BO and Bayesian Quadrature (BQ) is proposed for numerical implementation; upon incorporating the measurements obtained at the experimental design, BMU is implemented, with a specifically designed BQ algorithm, for probabilistic calibration of model parameters. The BOED and BMU are implemented alternately and sequentially until the global optimum of the physical system is reached with desired accuracy. Throughout the entire computational process, training data are reused and enriched via active learning strategies, leading to high computational efficiency; in addition, the proposed framework inherits the global convergence property of the traditional BO algorithm. The developments are ultimately demonstrated with numerical and PDE-constrained optimization problems.

Keywords: Bayesian optimization, Bayesian optimal experiment design, Bayesian model updating, Bayesian quadrature, Global optimization

1. Introduction

Bayesian Optimization (BO) is a probabilistic surrogate based active learning method for efficiently optimizing expensive-to-evaluate black-box functions with as few function evaluations as possible [1]. It has been widely applied across diverse fields of scientific research and technological development, where

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prominent practical benefits have been achieved. For example, in synthetic chemistry, BO has been utilized for reaction optimization, and extensive benchmark studies demonstrated its superiority, in terms of both efficiency and consistency, over traditional human decision-making strategy [2]; for material discovery, BO has been proved to be effective for guiding both experiments and computations [3]; in material design, a target-oriented BO scheme has been developed for navigating design configurations with target-specific properties with as few experimental iterations as possible [4, 5]; In drug discovery and clinical research, an automated and fully integrated workflow from per-table manufacturing to testing has been developed based on BO, and was demonstrated to be capable of substantially reducing drug R&D timeline [6]; BO has even been employed for optimizing the dessert recipes, yielding promising performance [7]. Beyond the above-mentioned examples, BO has also achieved substantial gains across many other areas such as petroleum exploitation and AI model training, further examples are omitted for brevity.

A typical BO algorithm hinges on two key components: probabilistic surrogate models and acquisition functions, where the former are employed for predicting the responses of the black-box functions with uncertainty quantification, for which Gaussian Process (GP) are Bayesian/ensembled Neural Network (BNN) are commonly suggested [8]; and the latter are responsible for sequentially acquiring training data to balance exploration and exploitation for rapid convergence to the global optimum [9]. During the past few decades, alternative acquisition functions have been developed, and typical examples include Expected Improvement (EI) function and its many variants [10], probability of improvement function [9], upper confidence bound function [11], knowledge gradient function [12], and predictive entropy search function [13]. Theoretical investigations have also been extensively presented for acquisition functions. For example, convergence rates have been theoretically investigated for different variants of EI functions (see e.g., [14, 15]). The aforementioned acquisition functions have also been extended for addressing more complex global optimization problems such as constrained optimization (see e.g., [16]) and multi-objective optimization (see e.g., [17]). As a summary, BO has evolved into a theoretically rigorous and fully mature computational framework, while practical engineering applications continue to expand rapidly.

In prevailing BO algorithms and their engineering practices, labels of training data are either acquired via experimental measurements or evaluated with black-box simulators. Nevertheless, in practice, both can be available, but the physical tests are extremely expensive, in terms of both economic expense and time consumption, to implement; while simulators may deliver extensive prediction uncertainties resulted from model simplification, unknown model parameters, etc. In this scenario, searching for the true global optimum at the lowest possible cost through the combination of physical tests and non-deterministic simulators, is a promising yet challenging task. A conceptually similar computational framework is the robust design optimization against model uncertainty, which aims to obtain the optimal design that is least sensitive to model uncertainty. Notably, this framework relies solely on information derived from the simulators whose parameters are non-deterministic (see e.g., [18, 19]).

We present Simulation-Test Integrated Global Optimization (STIGO), a computational framework for searching global optimum via automatic and dynamic fusion of physical tests and simulators, with proper treatments of both measurement noises and model uncertainties. It involves sequentially implementing Bayesian Optimal Experiment Design (BOED), for searching the design point which is most informative for global optimum under model uncertainty, and then Bayesian Model Updating (BMU), for probabilistic calibration of the model parameters by fusing the experimental observations generated from physical test implemented at the design point, until the global optimum is generated with desired accuracy. For BOED, a utility function is formulated, inspired by the EI function, for informing the optimal design, and a Bayesian computational method, which combines BO and Bayesian Quadrature (BQ), is developed for numerical search of this optimal experimental design (OED) with high efficiency and guarantee of convergence; while for BMU, another BQ rule is formulated for efficiently estimating the posterior density and producing the corresponding sample populations, which serve as inputs to BOED.

The rest of this work is organized as follows. Section 2 presents the STIGO framework, including the utility function for BOED and the mathematical formulation of the BMU problem; Section 3 concerns numerical implementation of STIGO, including two Bayesian numerical algorithms for addressing BOED and BMU sub-tasks respectively, as well as the complete numerical implementation procedures. Benchmark studies are presented in section 4 for demonstrating the effectiveness of the proposed STIGO framework, followed by conclusions, discussions and prospects in section 5.

2. Formulation of Simulation-Test Integrated Bayesian Optimization

Given the physical system under consideration, let $\boldsymbol{x} = (x_1, x_2, \cdots, x_d)$ denote the d-dimensional design vector, and y the output which is assumed to be scalar for simplicity. Suppose the dependence of y on $\boldsymbol{x}$ is described by a deterministic black-box function $y = \mathcal{P}(\boldsymbol{x})$, which is unknown. In practice, for example, $\boldsymbol{x}$ can represent the manufacturing process parameters, while y indicates the performance indicator of products. The task is to seek the optimal design $\boldsymbol{x}^*$ of the process parameters $\boldsymbol{x}$ that produces the best-performance product by sequentially querying y at a sequence of $\boldsymbol{x}$. The problem is mathematically formulated as:

$$\boldsymbol{x}^* = \text{argmax}_{\boldsymbol{x}\in\mathbb{X}}\mathcal{P}(\boldsymbol{x}) \tag{1}$$

, where $\mathbb{X} \subset \mathbb{R}^d$ indicates the feasibility region of $\boldsymbol{x}$. In the above setting, the label y at each $\boldsymbol{x}$ is obtained by measurement from an physical test, and is thus named as test-based BO.

On the other hand, suppose the objective function $\mathcal{P}(\boldsymbol{x})$ can also be predicted with an expensive-to-evaluate black-box simulator, denoted as $\mathcal{M}(\boldsymbol{x}, \boldsymbol{\theta})$, with no model discrepancy, where $\boldsymbol{\theta} = (\theta_1, \theta_2, \cdots, \theta_m)$ is the m-dimensional vector of model parameters. In case the value of $\boldsymbol{\theta}$ is exactly known, one can query the value of y at each $\boldsymbol{x}$ by calling this simulator, and the resultant scheme is named as model-based BO.

In a practical scenario, it often appears that physical tests and simulators are both available, but the physical test is extremely expensive to implement in terms of economy and/or time; and the model parameters $\boldsymbol{\theta}$ are deterministic but unknown, giving rise to uncertainty of model predictions. The aim of this work is to develop a methodology framework, which combines the physical tests and the non-deterministic simulators, for searching the global maximum point $\boldsymbol{x}^*$ with the least cost. The proposed framework, named STIGO, is schematically illustrated in Figure 1. Conceptually, it sequentially performs BOED and BMU, to produce a sequence of OED points $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_t$ and a corresponding sequence of experimental measurements $\mathrm{y}_1, \mathrm{y}_2, \cdots \mathrm{y}_t$, for approaching $\boldsymbol{x}^*$ by adaptively eliminating the model uncertainty for informing $\boldsymbol{x}^*$.

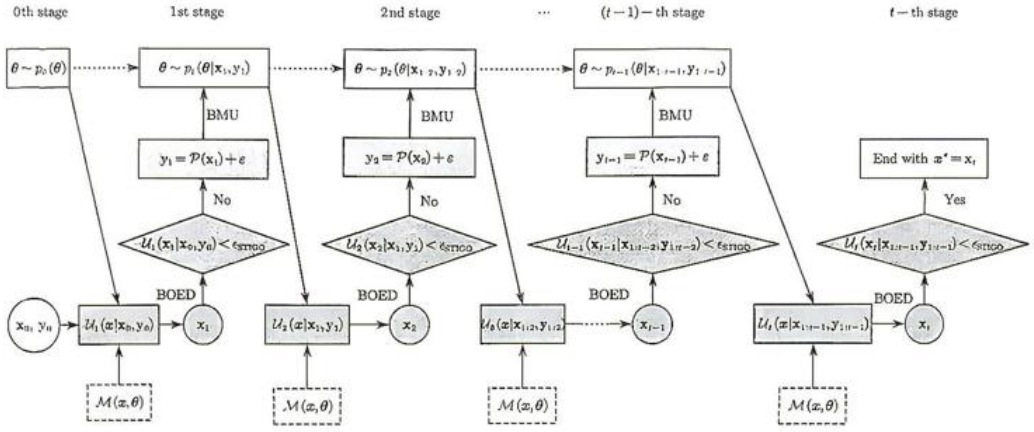


Figure 1: Schematically illustration of the proposed STIGO framework.

In case the response value is generated by calling the simulator $\mathcal{M}(\boldsymbol{x}, \boldsymbol{\theta})$ (required in the BOED and BMU procedures of Figure 1), the prediction definitely conveys uncertainty, resulted from the unknowness of the model parameters $\boldsymbol{\theta}$, which, before implementing any experimental test, is represented by a prior probability density $p_0(\boldsymbol{\theta})$. Alternatively, in case it is generated by implementing a physical test at any OED $\mathbf{x}_t$ (required between BOED and BMU procedures in Figure 1), the corresponding measurement of system response is given by:

$$\mathrm{y}_t = \mathcal{P}(\mathbf{x}_t) + \varepsilon \tag{2}$$

, where ε is the measurement error, assumed to be a Gaussian white noise with variance σ_{n}^2, i.e., $\varepsilon \sim \mathcal{N}(0, \sigma_{\mathrm{n}}^2)$.

Suppose an initial guess $\mathbf{x}_0$ of the optimal design is obtained, with its objective label value denoted as y_0. Then, initialized with $(\mathbf{x}_0, \mathrm{y}_0)$, the inference is implemented sequentially. At each step t, two key procedures need to be implemented. The first procedure implements the BOED to generate a design $\mathbf{x}_t$ which is most informative for the optimal design $\boldsymbol{x}^*$. Then, after generating the observation y_t by implementing a physical test at $\mathbf{x}_t$, the second procedure runs BMU to update the density of $\boldsymbol{\theta}$, this way to improve the belief on inferring the optimal design $\boldsymbol{x}^*$.

In a BOED procedure, a utility function requires to be developed for informing the most valuable design $\mathbf{x}_t$, by fusing three components which include, first, the time-to-date knowledge on $\boldsymbol{\theta}$ described by the conditional density $p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right)$, with $\mathbf{x}_{1:t-1}$ and $\mathbf{y}_{1:t-1}$ being the OEDs and the corresponding measurements up to the $(t-1)$-th stage; second, the information of measurement noise; and third, the computational model with parameter uncertainty. Inspired by the EI acquisition function originally developed for BO [9], the utility function is defined as:

$$\mathcal{U}_t\left(\boldsymbol{x}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right) = \int_{\mathbb{R}}\int_{\mathbb{T}} \max\left(\mathcal{M}\left(\boldsymbol{x}, \boldsymbol{\theta}\right) - y_{\max} + \varepsilon, 0\right) p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right) \phi\left(\varepsilon|0, \sigma_{\mathrm{n}}^2\right) \mathrm{d}\boldsymbol{\theta}\mathrm{d}\varepsilon \tag{3}$$

, where $y_{\max}$ indicates the reference value computed as the maximum observation up to the $(t-1)$-th stage, i.e., $y_{\max} = \max\left(\mathbf{y}_{1:t-1}\right)$, $\phi\left(\varepsilon|0, \sigma_{\mathrm{n}}^2\right)$ is the density of the Gaussian white noise related to $y_{\max}$, $\mathbf{x}_{1:t-1}$ and $\mathbf{y}_{1:t-1}$ refer to the OEDs of $\boldsymbol{x}$ and the corresponding experimental observations up to the $(t-1)$-th stage. Intuitively, this utility function measures the expected improvement of the objective function value at $\boldsymbol{x}$ compared to the reference value $y_{\max}$ observed experimentally up to the $(t-1)$-th stage, where the "expectation" is taken by integrating out the uncertainties of both $\boldsymbol{\theta}$ and ε. Then, the OED point for the t-th stage is computed by:

$$\mathbf{x}_t = \mathrm{argmax}_{\boldsymbol{x}\in\mathbb{X}} \mathcal{U}_t\left(\boldsymbol{x}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right). \tag{4}$$

The utility function defined by Eq. (3) is absolutely different from those developed for general BOED tasks which aim at effectively reducing the uncertainties associated with model parameters and/or model predictions (see, e.g., [20, 21, 22] for example). Notably, the one defined by Eq. (3) is specifically designed for STIGO, and is computationally much more tractable than those traditional ones defined with, e.g., Kullback-Leibler divergence or Wasserstein distance [23, 24, 21]. It should also be noted that, although the above utility function is inspired by the EI acquisition function, it does not admit a closed-form expression unless the stochastic process $\mathcal{M}\left(\boldsymbol{x}, \boldsymbol{\theta}\right)$ indexed by $\boldsymbol{\theta}$ is Gaussian. It is required to develop a cost-effective numerical algorithm for solving Eq. (4), which is nontrivial as it is a double-loop optimization task and the computational model $\mathcal{M}$ is generally expensive to evaluate.

After generating the observation $\mathbf{y}_t$ by implementing the physical test at $\mathbf{x}_t$, the BMU problem for the t-th step can be formulated. Specifically, the posterior density $p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right)$ generated at the $(t-1)$-th step is employed as the prior density. The likelihood function, based on the Gaussian white noise assumption, is formulated by $\mathcal{L}\left(\mathbf{x}_t, \mathbf{y}_t|\boldsymbol{\theta}\right) = \phi\left(\mathbf{y}_t - \mathcal{M}\left(\mathbf{x}_t, \boldsymbol{\theta}\right)|0, \sigma_{\mathrm{n}}^2\right)$. According to the Bayes' formula, the posterior density of model parameters for the t-th stage is formulated as:

$$p_t\left(\boldsymbol{\theta}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}\right) = \frac{\mathcal{L}\left(\mathbf{x}_t, \mathbf{y}_t|\boldsymbol{\theta}\right) p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right)}{Z_t} \tag{5}$$

, with $Z_t = \int_{\mathbb{T}} \mathcal{L}\left(\mathbf{x}_t, \mathbf{y}_t|\boldsymbol{\theta}\right) p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right) \mathrm{d}\boldsymbol{\theta}$ being the normalizing constant or model evidence. Esti-

mating posterior density $p_t\left(\boldsymbol{\theta}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}\right)$ or generating posterior samples is the core numerical task for BMU, which is non-trivial as each call of likelihood requires an evaluation of the expensive-to-evaluate model $\mathcal{M}$.

Finally, a stopping condition requires to be formulated to complete STIGO. Suppose for one step t, as shown in Figure 1, it appears that the following condition is reached for the OED point $\mathbf{x}_t$:

$$\mathcal{U}_t\left(\mathbf{x}_t|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right) < \epsilon_{\text{STIGO}} \tag{6}$$

, with ϵ_{STIGO} being a sufficiently small error tolerance, it means that the maximum value of system response can no longer be improved by collecting more test data, and thus the global maximum, denoted as $\boldsymbol{x}^* = \mathbf{x}_t$, is reached with high credibility.

In general, the numerical tasks for each stage of STIGO include searching $\mathbf{x}_t$ with global convergence, evaluating the value of $\mathcal{U}_t$ at $\mathbf{x}_t$ with desired accuracy, estimating the density p_t with given error tolerance and generating samples following p_t.

3. Numerical Implementation

In this section, numerical implementation of STIGO is concerned. For brevity, we denote $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{\theta})$ in case discrimination of $\boldsymbol{x}$ and $\boldsymbol{\theta}$ is not required. The algorithm is based on training a GP surrogate model of the computational model, which will be initialized with a set of N_0 training points, denoted as $\mathcal{D}_{N_0} = \left\{\left(\boldsymbol{z}^{(j)}, y^{(j)}\right)\right\}_{j=1}^{N_0}$ with $\boldsymbol{z}^{(j)} = \left(\boldsymbol{x}^{(j)}, \boldsymbol{\theta}^{(j)}\right)$. Practically, $\mathcal{D}_{N_0}$ can be generated using e.g., the Latin-hypercube sampling (LHS) scheme with $\boldsymbol{x}$ following uniform distribution within $\mathbb{X}$ and $\boldsymbol{\theta}$ following the prior distribution p_0, and then evaluating the computational model $y^{(j)} = \mathcal{M}\left(\boldsymbol{x}^{(j)}, \boldsymbol{\theta}^{(j)}\right)$. The value of y_0 can then be simply specified as $\mathrm{y}_0 = \max_{j=1}^{N_0} y^{(j)}$. With the above information as inputs, we need to develop two algorithms for respectively addressing the BOED problem formulated with Eq. (4) and the BMU problem formulated by Eq. (5). Hereafter in this section, we denote $p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right)$ by $p_{t-1}\left(\boldsymbol{\theta}\right)$, and $\mathcal{U}_t\left(\boldsymbol{x}|\mathbf{x}_{1:t-1}, \mathbf{y}_{1:t-1}\right)$ by $\mathcal{U}_t\left(\boldsymbol{x}\right)$ for conceptual simplicity. Further, we assume that a sample matrix $\mathbf{T}_{t-1} = \left\{\boldsymbol{\theta}_{t-1}^{(k)}\right\}_{k=1}^{M}$ following p_{t-1} is readily generated, which, for $t = 1$, is produced by LHS following p_0, and for $t > 1$, is created by the BMU procedure.

3.1. Collaborative and adaptive Bayesian optimization for BOED

Integrating ε in Eq. (3) out, the utility function can be reformulated as:

$$\mathcal{U}_t\left(\boldsymbol{x}\right) = \mathbb{E}_{p_{t-1}}\left[u\left(\boldsymbol{z}\right)\right] \tag{7}$$

, where $\mathbb{E}_{p_{t-1}}[\cdot]$ is the expectation operator w.r.t. p_{t-1} for integrating $\boldsymbol{\theta}$ out, and

$$u(\boldsymbol{z}) = (\mathcal{M}(\boldsymbol{z}) - y_{\max})\,\Phi\left(\frac{\mathcal{M}(\boldsymbol{z}) - y_{\max}}{\sigma_{\mathrm{n}}}\right) + \sigma_{\mathrm{n}}\phi\left(\frac{\mathcal{M}(\boldsymbol{z}) - y_{\max}}{\sigma_{\mathrm{n}}}\right) \tag{8}$$

, with $\Phi(\cdot)$ and $\phi(\cdot)$ indicating the standard Gaussian distribution and density functions respectively. One can refer to Appendix Appendix A for mathematical derivation of Eqs. (7) and (8), and Appendix Appendix B for mathematical properties of the u-function, i.e., the non-negativity and monotonicity w.r.t. the model $\mathcal{M}$. These two properties form the foundation for developing the effective Bayesian algorithm in this subsection.

With the above reformulation, the BOED problem can be mathematically explained as a double-loop numerical problem, with the outer loop for searching for the maximizer of the utility function over the feasibility space of $\boldsymbol{x}$; and for each iteration point of $\boldsymbol{x}$, the inner loop for evaluating the integral of the function $u(\boldsymbol{z})$ weighted by the density p_{t-1}. For this type of nested problem, a theoretically sounding method is the Collaborative and Adaptive Bayesian Optimization (CABO) [25], as it provides guarantees of global convergence for both loops and is highly efficient. The basic idea of CABO is to actively learn a GP surrogate for, e.g., $u(\boldsymbol{z})$, in the augmented space of $\boldsymbol{z}$, allowing for sequentially performing BO in the marginal space of $\boldsymbol{x}$ and then BQ in the marginal space of $\boldsymbol{\theta}$, this way to find the global optimal point of $\boldsymbol{x}$ and meanwhile, evaluate the value of the utility function at this point with desired accuracy. Unfortunately, the original version of CABO reported in Ref. [25] is tested to be ineffective for this specific task, as the values of the u- and $\mathcal{U}$-functions vanish numerically in most regions (although their values rarely equal exactly to zero), especially in case the reference value $y_{\max}$ approaches the global optimum, making it challenging to be approximated with a GP model. Besides, it is found that the gradients of both u- and $\mathcal{U}$-functions over each x_i also tend to vanish numerically in majority regions, making the gradient-based algorithms inapplicable. A typical example of the u- and $\mathcal{U}$-functions as well as their gradients are presented in Figure 2 for illustrating the above challenges. In what follows, we develop a new CABO algorithm, tailored to the above features of u- and $\mathcal{U}$-functions, for efficiently searching the OED point $\mathbf{x}_t$.

Given training data $\mathcal{D}_N = \left\{\left(\boldsymbol{z}^{(k)}, y^{(k)}\right)\right\}_{k=1}^{N}$, a GP surrogate model $\mathcal{M}_N(\boldsymbol{z}) \sim \mathcal{GP}\left(\mu_{\mathcal{M}}(\boldsymbol{z}), c_{\mathcal{M}}(\boldsymbol{z}, \tilde{\boldsymbol{z}})\right)$, can be trained, where $\mu_{\mathcal{M}}(\boldsymbol{z})$ is the mean prediction and $c_{\mathcal{M}}(\boldsymbol{z}, \tilde{\boldsymbol{z}})$ refers to the prediction covariance. The prediction variance, computed as $\sigma^2_{\mathcal{M}}(\boldsymbol{z}) = c_{\mathcal{M}}(\boldsymbol{z}, \boldsymbol{z})$, serves as a quantification of the prediction uncertainty. Mathematically, they are formulated as:

$$\mu_{\mathcal{M}}(\boldsymbol{z}) = m(\boldsymbol{z}) + \boldsymbol{\kappa}(\boldsymbol{z}, Z)\,\mathbf{K}^{-1}(\mathcal{Y} - m(Z)), \tag{9a}$$

$$c_{\mathcal{M}}(\boldsymbol{z}, \tilde{\boldsymbol{z}}) = \boldsymbol{\kappa}(\boldsymbol{z}, \tilde{\boldsymbol{z}}) - \boldsymbol{\kappa}(\boldsymbol{z}, Z)\,\mathbf{K}^{-1}\boldsymbol{\kappa}(Z, \tilde{\boldsymbol{z}}) \tag{9b}$$

, where Z is the matrix with its k-th row being $\boldsymbol{z}^{(k)}$, $\mathcal{Y}$ indicates the column vector with its k-th element

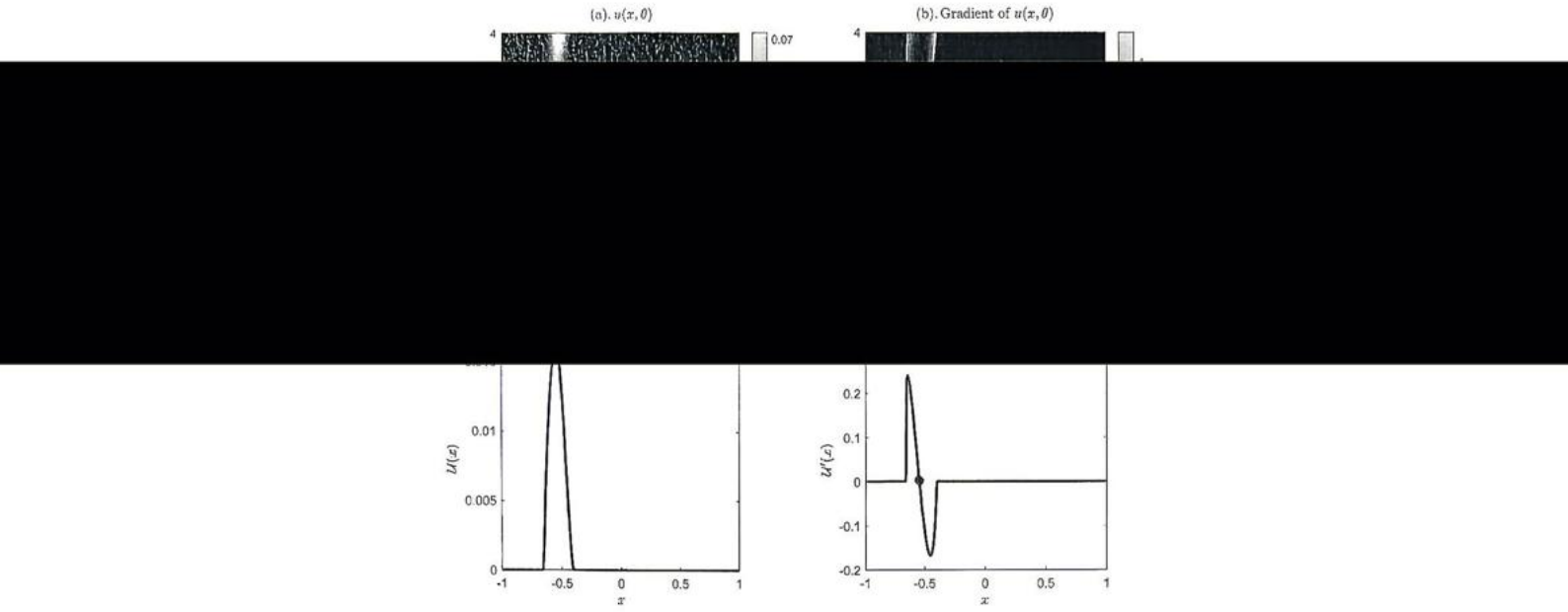


Figure 2: An example of the u- and $\mathcal{U}$-functions and their gradients.

being $y^{(k)}$, $m(\boldsymbol{z})$ denotes the prior mean function which can be assumed as zero, constant, or polynomials, $\kappa(\boldsymbol{z}, \tilde{\boldsymbol{z}})$ is the covariance function which can be assumed to be any kernel functions like squared exponential or Matérn kernel, $\boldsymbol{\kappa}(\boldsymbol{z}, \mathcal{Z})$ indicates a row vector with its j-th element being $\kappa\left(\boldsymbol{z}, \boldsymbol{z}^{(j)}\right)$, and $\mathbf{K}$ is the Gram matrix with its (j,k)-th element being $\kappa\left(\boldsymbol{z}^{(j)}, \boldsymbol{z}^{(k)}\right)$. Following the Gaussianity of $\mathcal{M}_N$, a Credible Bounds (CIs) of $\mathcal{M}_N$ can be formulated as $\left[\underline{\mathcal{M}}, \overline{\mathcal{M}}\right]$, with

$$\underline{\mathcal{M}} = \mu_{\mathcal{M}} - \beta\sigma_{\mathcal{M}} \tag{10a}$$

$$\overline{\mathcal{M}} = \mu_{\mathcal{M}} + \beta\sigma_{\mathcal{M}} \tag{10b}$$

, where β corresponds to the Z-score, which is determined by the credible level, and controls the trade-off between exploration and exploitation of CABO. As the u-function defined by Eq. (8) is monotonically increasing with respect to $\mathcal{M}$ (see Appendix Appendix B for proof), a corresponding lower bound $\underline{u}$, upper bound $\overline{u}$ and mean estimate μ_u of u can be generated by replacing $\mathcal{M}$ in Eq. (8) with $\underline{\mathcal{M}}$, $\overline{\mathcal{M}}$ and $\mu_{\mathcal{M}}$ respectively.

Each iteration step of CABO consists of a BO inference and a BQ inference. For BO procedure, an

acquisition function, named Upper Credible Bound (UCB) function, is formulated and estimated by:

$$\mathcal{A}_{\mathrm{UCB}}(\boldsymbol{x}) = \int_{\mathbb{T}} \overline{u}(\boldsymbol{x}, \boldsymbol{\theta}) p_{t-1}(\boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta} \cong \frac{1}{M} \sum_{k=1}^{M} \overline{u}\left(\boldsymbol{x}, \boldsymbol{\theta}_{t-1}^{(k)}\right) \tag{11}$$

, which is equivalent to the traditional UCB function in BO (see e.g., [11]). A larger value of the UCB function corresponds to a higher posterior mean prediction of the u-function and/or greater predictive uncertainty, where the former governs exploitation, and the latter drives exploration. Consequently, the optimal training point of $\boldsymbol{x}$ is specified by maximizing the UCB function, i.e.,

$$\boldsymbol{x}^{+} = \mathrm{argmax}_{\boldsymbol{x} \in \mathbb{X}} \mathcal{A}_{\mathrm{UCB}}(\boldsymbol{x}). \tag{12}$$

As the UCB function admits a closed-form estimator which is trivial to evaluate, the above optimization problem can be efficiently addressed with any heuristic algorithms, like Particle Swarm Optimization (PSO) and Genetic Algorithm (GA).

Following $\boldsymbol{x}^{+}$ being generated, a joint training point $\boldsymbol{\theta}^{+}$ needs to be specified for the BQ procedure, with the aim of improving the prediction accuracy of $\mathcal{U}_t(\boldsymbol{x}^{+})$ to a greatest extent. To this aim, the following acquisition function, named Prediction Uncertainty Quantification (PUQ) function, is formulated:

$$\mathcal{A}_{\mathrm{PUQ}}(\boldsymbol{\theta} | \boldsymbol{x}^{+}) = \overline{u}(\boldsymbol{x}^{+}, \boldsymbol{\theta}) - \underline{u}(\boldsymbol{x}^{+}, \boldsymbol{\theta}). \tag{13}$$

As $\overline{\mathcal{U}}_t(\boldsymbol{x}^{+}) - \underline{\mathcal{U}}_t(\boldsymbol{x}^{+}) = \int_{\mathbb{T}} \mathcal{A}_{\mathrm{PUQ}}(\boldsymbol{\theta} | \boldsymbol{x}^{+}) p_{t-1}(\boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta}$ quantifies the prediction uncertainty of $\mathcal{U}_t(\boldsymbol{x}^{+})$, the sample point in $\mathbf{T}_{t-1}$ with the highest PUQ value is the one whose GP prediction uncertainty contributes the most to that of $\mathcal{U}(\boldsymbol{x}^{+})$. Thereof, the new training point of $\boldsymbol{\theta}$ can be specified as:

$$\boldsymbol{\theta}^{+} = \mathrm{argmax}_{\boldsymbol{\theta} \in \mathbf{T}_{t-1}} \mathcal{A}_{\mathrm{PUQ}}(\boldsymbol{\theta} | \boldsymbol{x}^{+}) \tag{14}$$

, where $\boldsymbol{\theta}^{+}$ is selected from $\mathbf{T}_{t-1}$. As the PUQ function admits a closed-form expression, the cost for specifying $\boldsymbol{\theta}^{+}$ is extremely low.

With the above two acquisition functions, a new training point $\boldsymbol{z}^{+} = (\boldsymbol{x}^{+}, \boldsymbol{\theta}^{+})$ is produced. Further, with the response value $y^{+} = \mathcal{M}(\boldsymbol{z}^{+})$ being computed by calling the simulator, the new data $(\boldsymbol{z}^{+}, y^{+})$ is used for enriching the training data $\mathcal{D}_N$ and updating the GP model to continue to implement BO and BQ sequentially until the global maximum point of $\mathcal{U}(\boldsymbol{x})$ is identified with desired accuracy. It is then necessary to develop a stopping condition for indicating if the above target is reached. Specifically, for BO procedure,

the stopping condition is defined as:

$$\frac{|\mathcal{A}_{\mathrm{UCB}}(x^+) - \mathcal{A}_{\mathrm{UCB},N-1}|}{\mathcal{A}_{\mathrm{UCB}}(x^+)} < \epsilon_{\mathrm{BO}} \tag{15}$$

, where $\mathcal{A}_{\mathrm{UCB},N-1}$ indicates the largest UCB value computed and recorded in the last iteration step, which can be set as $y_{\max}$ for the first iteration step, and ϵ_{BO} refers to the threshold for indicating the error tolerance of searching $\mathbf{x}_t$. For BQ procedure, the stopping condition is formulated as:

$$\frac{\overline{\mathcal{U}}_t(x^+) - \underline{\mathcal{U}}_t(x^+)}{\overline{\mathcal{U}}_t(x^+) + \underline{\mathcal{U}}_t(x^+)} < \epsilon_{\mathrm{BQ}} \tag{16}$$

, where ϵ_{BQ} is the threshold for indicating the error tolerance of estimating the integral equation $\mathcal{U}_t(x^+) = \int_{\mathbb{T}} u(x^+, \boldsymbol{\theta}) p_{t-1}(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta}$, $\overline{\mathcal{U}}_t(x^+)$ and $\underline{\mathcal{U}}_t(x^+)$ are the upper and lower credible bounds for estimating $\mathcal{U}_t(x^+)$, and are estimated by:

$$\overline{\mathcal{U}}_t(x^+) \cong \frac{1}{M} \sum_{k=1}^{M} \overline{u}\left(x^+, \boldsymbol{\theta}_{t-1}^{(k)}\right) \tag{17a}$$

$$\underline{\mathcal{U}}_t(x^+) \cong \frac{1}{M} \sum_{k=1}^{M} \underline{u}\left(x^+, \boldsymbol{\theta}_{t-1}^{(k)}\right). \tag{17b}$$

To summary, in case both Eqs. (15) and (16) are satisfied, the global maximum point $\mathbf{x}_t$ is found with desired accuracy, and its utility function value $\mathcal{U}_t(\mathbf{x}_t)$ is estimated with given error tolerance. The mean estimate of $\mathcal{U}_t(\mathbf{x}_t)$, formulated as:

$$\mu_{\mathcal{U}_t}(\mathbf{x}_t) \cong \frac{1}{M} \sum_{k=1}^{M} \mu_u\left(\mathbf{x}_t, \boldsymbol{\theta}_{t-1}^{(k)}\right) \tag{18}$$

, will then be utilized for indicating the convergence of STIGO, as indicated by Eq. (6).

With the above theoretical developments ready, the pseudocode of CABO for searching $\mathbf{x}_t$ is summarized in Algorithm 1. The inputs to this algorithm include the model function $\mathcal{M}$, the measurement noise variance σ_{n}^2, the prior density p_{t-1} which is produced by the BMU procedure for $t > 1$ and equals to p_0 for $t = 0$, the sample population $\mathbf{T}_{t-1}$ following p_{t-1} generated by the BMU procedure for $t > 1$ or by LHS following p_0 for $t = 1$, the training data $\mathcal{D}_N$, and the reference objective function value $y_{\max}$ which is computed as the maximum measurement thus far. The outputs of Algorithm 1 include the OED point $\mathbf{x}_t$, the mean estimate $\mu_{\mathcal{U}_t}(\mathbf{x}_t)$ of the utility function at $\mathbf{x}_t$, as well as the enriched training data $\mathcal{D}_N$, where $\mu_{\mathcal{U}_t}(\mathbf{x}_t)$ will be used in subsequent procedure to indicate the global stopping condition of STIGO.

The parameters required to be pre-specified for Algorithm 1 include the sample size M (suggested to be $5 \sim 10 \times 10^3$), the Z-score β (suggested to be 1), the stopping threshold ϵ_{BO} (suggested to be $1 \sim 5 \times 10^{-3}$), and ϵ_{BQ} (depends on β, and suggested to be 0.02~0.05 for $\beta = 1$), as well as the maximum allowable

number $N_{\lim}$ of model evaluations (if required by users).

Algorithm 1: The CABO algorithm for BOED inference in the t-th step.

Input: model $\mathcal{M}$, noise variance σ_{n}^2, prior density p_{t-1}, sample matrix $\mathbf{T}_{t-1}$ following p_{t-1}, training data $\mathcal{D}_N$, reference value $y_{\max}$.

Output: OED $\mathbf{x}_t$, mean estimate $\mu_{\mathcal{U}_t}(\mathbf{x}_t)$, and enriched training data $\mathcal{D}_N$.

Parameters: Sample size M, Z-score β, thresholds ϵ_{BO} and ϵ_{BQ}, allowed model calls $N_{\lim}$.

$\mathcal{A}_{\mathrm{UCB},N-1} \leftarrow y_{\max}$;
while $N \leqslant N_{\lim}$ **do**
  $\mathcal{M}_N(\boldsymbol{z}) \leftarrow$ train or update GP model with $\mathcal{D}_N$;
  $\overline{\mathcal{M}}(\boldsymbol{z}), \underline{\mathcal{M}}(\boldsymbol{z}) \leftarrow$ compute by Eq. (10);
  $\overline{u}(\boldsymbol{z})$, $\underline{u}(\boldsymbol{z}) \leftarrow$ compute by replacing $\mathcal{M}$ in Eq. (8) with $\overline{\mathcal{M}}$ and $\underline{\mathcal{M}}$ respectively;
  $\boldsymbol{x}^+ \leftarrow$ maximize $\mathcal{A}_{\mathrm{UCB}}(\boldsymbol{x})$ over $\mathbb{X}$ using PSO/GA, with $\mathcal{A}_{\mathrm{UCB}}(\boldsymbol{x})$ estimated by Eq. (11);
  $\mathcal{A}_{\mathrm{UCB},N} \leftarrow \mathcal{A}_{\mathrm{UCB}}(\boldsymbol{x}^+)$;
  $\boldsymbol{\theta}^+ \leftarrow$ search from $\mathbf{T}_{t-1}$ the point with the highest $\mathcal{A}_{\mathrm{PUQ}}(\boldsymbol{\theta}|\boldsymbol{x}^+)$ value, with $\mathcal{A}_{\mathrm{PUQ}}(\boldsymbol{\theta}|\boldsymbol{x}^+)$ computed using Eq. (13);
  $\overline{\mathcal{U}}_t(\boldsymbol{x}^+)$, $\underline{\mathcal{U}}_t(\boldsymbol{x}^+) \leftarrow$ compute by Eq. (17);
  if *Both Eqs.* (15) *and* (16) *satisfied* **then**
    $\mathbf{x}_t \leftarrow \boldsymbol{x}^+$;
    break **while**.
  else
    $y^+ \leftarrow \mathcal{M}(\boldsymbol{z}^+)$ with $\boldsymbol{z}^+ = (\boldsymbol{x}^+, \boldsymbol{\theta}^+)$, $N \leftarrow N+1$;
    $\mathcal{D}_N \leftarrow \mathcal{D}_{N-1} \cup \{(\boldsymbol{z}^+, y^+)\}$.
$\mu_u(\boldsymbol{z}) \leftarrow$ compute by replacing $\mathcal{M}$ in Eq. (8) with its mean prediction $\mu_{\mathcal{M}}$;
$\mu_{\mathcal{U}_t}(\mathbf{x}_t) \leftarrow$ compute using Eq. (18).

3.2. Bayesian quadrature for BMU

Given $\mathbf{x}_t$ being designed with BOED, and the measurement, denoted as y_t, being obtained by implementing the physical test at $\mathbf{x}_t$, it is then required to implement the BMU procedure for updating the density of $\boldsymbol{\theta}$. The numerical aim of this procedure is to estimate the posterior density formulated with Eq. (5), and possibly also the model evidence Z_t, with desired accuracy and as few simulator calls as possible. We develop a specifically designed BQ method to fill this gap. Alternative BQ rules have been developed, most of which are formulated by approximating the logarithm of likelihood with a GP model [26, 27, 28], while others are based on training a GP surrogate for the model $\mathcal{M}$ [29, 30]. We use the latter treatment but propose new inference scheme to allow for reusing the training data produced by the BOED procedure and acquiring effective training data.

Denote the training data produced by BOED as $\mathcal{D}_N$, with which, a GP model $\mathcal{M}_N(\boldsymbol{z})$ is trained. A

stochastic process surrogate model can then be formulated for approximating the likelihood function as:

$$\mathcal{L}_N\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right) = \phi\left(\mathrm{y}_t - \mathcal{M}_N\left(\mathbf{x}_t, \boldsymbol{\theta}\right)|0, \sigma_{\mathrm{n}}^2\right) \tag{19}$$

, which is definitely non-Gaussian. The induced random variable,

$$Z_{t,N} = \int_{\mathbb{T}} \mathcal{L}_N\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right) p_{t-1}\left(\boldsymbol{\theta}\right) \mathrm{d}\boldsymbol{\theta} \tag{20}$$

, used for predicting the model evidence Z_t, is also non-Gaussian. Fortunately, the posterior mean $\mu_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right)$, variance $\sigma_{\mathcal{L}}^2\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right)$ and covariance $c_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}\right)$ of the probabilistic surrogate $\mathcal{L}_N\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right)$ all admit closed-form expressions (see Appendix Appendix C). The posterior mean and variance of $Z_{t,N}$ are then formulated as:

$$\mu_{Z_t} = \int_{\mathbb{T}} \mu_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right) p_{t-1}\left(\boldsymbol{\theta}\right) \mathrm{d}\boldsymbol{\theta} \tag{21a}$$

$$\sigma_{Z_t}^2 = \int_{\mathbb{T}\times\mathbb{T}} c_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}\right) p_{t-1}\left(\boldsymbol{\theta}\right) p_{t-1}\left(\tilde{\boldsymbol{\theta}}\right) \mathrm{d}\boldsymbol{\theta}\mathrm{d}\tilde{\boldsymbol{\theta}} \tag{21b}$$

, where $\sigma_{Z_t}^2$ quantifies the prediction uncertainty associated with estimating the model evidence Z_t using μ_{Z_t}. Then, the Posterior Variance Contribution (PVC) function, originally developed in Ref. [31] and then extended to BMU problem in Ref. [26], is utilized for adaptive design of training data, and in this context, it is formulated as:

$$\mathcal{A}_{\mathrm{PVC}}\left(\boldsymbol{\theta}\right) = \int_{\mathbb{T}} c_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}\right) p_{t-1}\left(\tilde{\boldsymbol{\theta}}\right) \mathrm{d}\tilde{\boldsymbol{\theta}}. \tag{22}$$

Mathematically, as $\sigma_{Z_t}^2 = \int_{\mathbb{T}} \mathcal{A}_{\mathrm{PVC}}\left(\boldsymbol{\theta}\right) p_{t-1}\left(\boldsymbol{\theta}\right) \mathrm{d}\boldsymbol{\theta}$ holds, the PVC function quantifies the contribution of the GP prediction uncertainty at the location $\left(\mathbf{x}_t, \boldsymbol{\theta}\right)$ to the prediction variance of $Z_{t,N}$, with the integration of the spatial correlation information of this point with the other points over the whole support of p_{t-1}.

Next, numerical estimators for μ_{Z_t}, $\sigma_{Z_t}^2$ and $\mathcal{A}_{\mathrm{PVC}}\left(\boldsymbol{\theta}\right)$ are concerned. Similar to the inputs to Algorithm 1, assume that a sample population $\mathbf{T}_{t-1} = \left\{\boldsymbol{\theta}_{t-1}^{(k)}\right\}_{k=1}^{M}$ have been generated following the distribution p_{t-1} from the BMU procedure at the $(t-1)$-th stage for $t > 1$, or by LHS following p_0 for $t = 1$. Further, an i.i.d. sample population $\tilde{\mathbf{T}}_{t-1} = \left\{\tilde{\boldsymbol{\theta}}_{t-1}^{(k)}\right\}_{k=1}^{M}$ is generated by random permuting the rows of $\mathbf{T}_{t-1}$. Then, the MC estimators for μ_{Z_t} and $\sigma_{Z_t}^2$ are formulated as:

$$\mu_{Z_t} \cong \frac{1}{M}\sum_{k=1}^{M} \mu_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}_{t-1}^{(k)}\right) \tag{23a}$$

$$\sigma_{Z_t}^2 \cong \frac{1}{M}\sum_{k=1}^{M} c_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}_{t-1}^{(k)}, \tilde{\boldsymbol{\theta}}_{t-1}^{(k)}\right). \tag{23b}$$

A MC estimator of the PVC function $\mathcal{A}_{\mathrm{PVC}}(\boldsymbol{\theta})$ is formulated as:

$$\mathcal{A}_{\mathrm{PVC}}(\boldsymbol{\theta}) \cong \frac{1}{M}\sum_{k=1}^{M} c_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}, \boldsymbol{\theta}_{t-1}^{(k)}\right). \tag{24}$$

The training point $\boldsymbol{\theta}^{+}$ is selected from $\mathbf{T}_{t-1}$ as the one with the highest PVC value:

$$\boldsymbol{\theta}^{+} = \mathrm{argmax}_{\boldsymbol{\theta}\in\mathbf{T}_{t-1}}\mathcal{A}_{\mathrm{PVC}}(\boldsymbol{\theta}). \tag{25}$$

By computing $y^{+} = \mathcal{M}\left(\mathbf{x}_t, \boldsymbol{\theta}^{+}\right)$, the GP model can be updated using the new training point $\left(\mathbf{x}_t, \boldsymbol{\theta}^{+}, y^{+}\right)$.

The stopping condition for the BMU procedure is formulated as:

$$\frac{\sigma_{Z_t}}{\mu_{Z_t}} < \epsilon_{\mathrm{BMU}} \tag{26}$$

, where ϵ_{BMU} is error tolerance parameter. Satisfaction of Eq. (26) indicates that the estimator μ_{Z_t} of Z_t achieves a required accuracy. Furthermore, as Z_t is the integral of the likelihood, the mean estimator $\mu_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right)$ of the likelihood also attains a desired accuracy. A mean estimator of the posterior density $p_t\left(\boldsymbol{\theta}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}\right)$ is presented as:

$$\mu_{p_t}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}\right) = \frac{\mu_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}\right) p_{t-1}\left(\boldsymbol{\theta}\right)}{\mu_{Z_t,N}} \triangleq p_t\left(\boldsymbol{\theta}\right) \tag{27}$$

, where $p_{t-1}\left(\boldsymbol{\theta}\right)$ can be given as its mean prediction $\mu_{p_{t-1}}\left(\boldsymbol{\theta}\right)$, generated in the last implementation of BMU procedure. The posterior density $\mu_{p_t}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}\right)$ formulated with Eq. (27) is then served as the prior density $p_t\left(\boldsymbol{\theta}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}\right)$ for the $(t+1)$ stage, and is abbreviated as $p_t\left(\boldsymbol{\theta}\right)$ for conceptual simplicity.

Given the mean estimator of Eq. (27) and the sample set $\mathbf{T}_{t-1}$, the Resampling-then-Metropolis-Hasting (R-MH) algorithm reported in Ref.[28, 32] can be utilized to produce the sample population $\mathbf{T}_t = \left\{\boldsymbol{\theta}_t^{(k)}\right\}_{k=1}^{M}$ which follows the posterior distribution $p_t\left(\boldsymbol{\theta}\right)$. The sample population $\mathbf{T}_t$ will then be utilized for the inference of the $(t+1)$-th stage. Specifically, for each sample $\boldsymbol{\theta}_{t-1}^{(k)} \in \mathbf{T}_{t-1}$, the weight is computed as :

$$w_t^{(k)} = \mu_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}_{t-1}^{(k)}\right). \tag{28}$$

The following discrete probability distribution law can be obtained:

$$\mathbb{P}\left\{\boldsymbol{\theta} = \boldsymbol{\theta}_{t-1}^{(k)}\right\} \sim \bar{w}_t^{(k)} = \frac{w_t^{(k)}}{\sum_{k=1}^{M} w_t^{(k)}} \tag{29}$$

, following which, a seed sample matrix $\bar{\mathbf{T}}_t = \left\{\bar{\boldsymbol{\theta}}_t^{(k)}\right\}_{k=1}^{M}$, of the same size, following the posterior distribution

$p_t(\boldsymbol{\theta})$ can be generated by sampling from $\mathbf{T}_{t-1}$ with replacement. Inevitably, $\bar{\mathbf{T}}_t$ contains a certain amount of repeated samples, which is referred to as sample degeneracy [33]. Moreover, in case the data evidence is strong, this issue can be so severe as to compromise the accuracy of subsequent inference. This is alleviated by growing a Markov chain with each sample $\bar{\boldsymbol{\theta}}_t^{(k)} \in \bar{\mathbf{T}}_t$ as the seed. The covariance of the candidate sample matrix $\bar{\mathbf{T}}_t$ can be estimated as:

$$\Sigma_{\boldsymbol{\theta},t} = \sum_{k=1}^{M} \bar{w}_t^{(k)} \left(\boldsymbol{\theta}_{t-1}^{(k)} - \bar{\boldsymbol{\theta}}_{t-1}\right)^{\top} \left(\boldsymbol{\theta}_{t-1}^{(k)} - \bar{\boldsymbol{\theta}}_{t-1}\right) \tag{30}$$

, where $\bar{\boldsymbol{\theta}}_{t-1} = \sum_{k=1}^{M} \boldsymbol{\theta}_{t-1}^{(k)} \bar{w}_t^{(k)}$ is the sample mean. This covariance will be used for formulating the proposal distribution for growing Markov chain. Specifically, with $\bar{\boldsymbol{\theta}}_t^{(k)} \in \bar{\mathbf{T}}_t$ as the seed $\bar{\boldsymbol{\theta}}_t^{(k,0)}$ of the k-th chain, the Metropolis-Hasting (MH) algorithm to produce the $(s+1)$-th state of this chain consists of the following two steps [28, 32]:

i. Generate a candidate sample $\boldsymbol{\theta}_{\text{cand}}$ following a proposal distribution which can be defined as, e.g., the Gaussian distribution with mean $\bar{\boldsymbol{\theta}}_t^{(k,s)}$ and covariance $\eta^2 \Sigma_{\boldsymbol{\theta},t}$, where $\eta = 0.2$ is a scaling parameter;

ii. Accept $\boldsymbol{\theta}_{\text{cand}}$ as $\bar{\boldsymbol{\theta}}_t^{(k,s+1)}$ with probability $\alpha = \min\left(1, p_t\left(\boldsymbol{\theta}_{\text{cand}}\right) / p_t\left(\bar{\boldsymbol{\theta}}_t^{(k,s)}\right)\right)$; if rejected, let $\bar{\boldsymbol{\theta}}_t^{(k,s+1)} = \bar{\boldsymbol{\theta}}_t^{(k,s)}$.

Repeat the above procedure to grow M chains, each with length L, and collect the last state $\bar{\boldsymbol{\theta}}_t^{(k,L)}$ of each chain as the target sample $\boldsymbol{\theta}_t^{(k)}$ to form the sample population $\mathbf{T}_t$, which definitely follows exactly p_t. The sample matrix $\mathbf{T}_t$ generated with the above R-MH scheme can effectively avoid the issues of sample degeneracy and sample correlation, which benefits the subsequent estimations formulated with these samples. Moreover, as the mean estimate of the posterior density, formulated with Eq. (27), admits a closed-form expression and requires no calls to the computational model, the acceptance probability α becomes trivial to compute, rendering the computational cost of the above R-MH scheme very low.

As a summary, the pseudocode of the BQ and R-MH algorithms developed for addressing the BMU issue are summarized in Algorithm 2. The inputs to this algorithm consist of the OED point $\mathbf{x}_t$ produced by Algorithm 1, the measurement y_t produced by implementing the physical test at $\mathbf{x}_t$, the noise variance σ_{n}^2, the sample matrix $\mathbf{T}_{t-1}$, and the prior density p_{t-1} which equals to p_0 (for $t = 1$) or produced by the last implementation of BMU. The outputs of Algorithm 2 consist of the mean estimate p_t of the posterior density and the mean estimate μ_{Z_t} of the model evidence Z_t, both with desired accuracy, the sample matrix $\mathbf{T}_t$ following p_t, as well as the training data $\mathcal{D}_N$ enriched with this algorithm.

The parameters of Algorithm 2 include the sample size (set to be the same as that for Algorithm 1), the chain length L (suggested to be 30$\sim$50), the scaling parameter η (suggested to be 0.2) and the stopping threshold ϵ_{BMU}(suggested to be $0.02 \sim 0.05$).

Algorithm 2: The BQ and R-MH sampling algorithms for BMU problem.

Input: OED $\mathbf{x}_t$, measurement y_t, noise variance σ_{n}^2, sample matrix $\mathbf{T}_{t-1}$, training data $\mathcal{D}_N$, prior density p_{t-1}.

Output: mean estimates p_t and Z_t, sample matrix $\mathbf{T}_t$, and the enriched training data $\mathcal{D}_N$.

Parameters: sample size M, chain length L, scaling parameter η and threshold ϵ_{BMU}.

$\tilde{\mathbf{T}}_{t-1} \leftarrow$ randomly permute the rows of $\mathbf{T}_{t-1}$;
$\mathcal{M}_N(\boldsymbol{z}) \leftarrow$ train a GP model with $\mathcal{D}_N$;
$\mu_{Z_t}, \sigma_{Z_t} \leftarrow$ compute by Eq. (23);
while *Eq. (26) is not satisfied* **do**
  $\boldsymbol{\theta}^+ \leftarrow$ search from $\mathbf{T}_{t-1}$ with highest $\mathcal{A}_{\mathrm{PVC}}(\boldsymbol{\theta})$ value, with $\mathcal{A}_{\mathrm{PVC}}(\boldsymbol{\theta})$ computed by Eq. (24);
  $y^+ \leftarrow \mathcal{M}(\mathbf{x}_t, \boldsymbol{\theta}^+)$, $N \leftarrow N+1$;
  $\mathcal{D}_N \leftarrow \mathcal{D}_{N-1} \cup \{(\mathbf{x}_t, \boldsymbol{\theta}^+, y^+)\}$;
  $\mathcal{M}_N(\boldsymbol{z}) \leftarrow$ update the GP model with $\mathcal{D}_N$;
  $\mu_{Z_t}, \sigma_{Z_t} \leftarrow$ compute by Eq. (23);
$p_t(\boldsymbol{\theta}) \leftarrow$ estimate $\mu_{p_t}(\boldsymbol{\theta}|\mathbf{x}_{1:t}, \mathrm{y}_{1:t})$ by Eq. (27);
$w_t^{(k)} \leftarrow \mu_{\mathcal{L}}\left(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}_{t-1}^{(k)}\right)$ for each $\boldsymbol{\theta}_{t-1}^{(k)} \in \mathbf{T}_{t-1}$;
$\bar{\mathbf{T}}_t = \left\{\bar{\boldsymbol{\theta}}_t^{(k)}\right\}_{k=1}^{M} \leftarrow$ sampling from $\mathbf{T}_{t-1}$ following the distribution law of Eq. (29);
$\Sigma_{\boldsymbol{\theta},t} \leftarrow$ compute by Eq. (30);
for $k = 1 : M$ **do**
  $\bar{\boldsymbol{\theta}}_t^{(k,0)} \leftarrow \bar{\boldsymbol{\theta}}_t^{(k)}$;
  for $s = 0 : L-1$ **do**
    $\boldsymbol{\theta}_{\mathrm{cand}} \leftarrow$ sampling from the Gaussian distribution with mean $\bar{\boldsymbol{\theta}}_t^{(k,s)}$ and covariance $\eta^2\Sigma_{\boldsymbol{\theta},t}$;
    $\alpha \leftarrow \min\left(1, p_t(\boldsymbol{\theta}_{\mathrm{cand}})/p_t\left(\bar{\boldsymbol{\theta}}_t^{(k,s)}\right)\right)$;
    if $\mathrm{rand}(0,1) < \alpha$ **then**
      $\bar{\boldsymbol{\theta}}_t^{(k,s+1)} \leftarrow \boldsymbol{\theta}_{\mathrm{cand}}$;
    else
      $\bar{\boldsymbol{\theta}}_t^{(k,s+1)} \leftarrow \bar{\boldsymbol{\theta}}_t^{(k,s)}$;
  $\boldsymbol{\theta}_t^{(k)} \leftarrow \bar{\boldsymbol{\theta}}_t^{(k,L)}$;
$\mathbf{T}_t \leftarrow \left\{\boldsymbol{\theta}_t^{(k)}\right\}_{k=1}^{M}$.

3.3. The complete implementation procedure

Given Algorithms 1 and 2 being well developed, the complete implementation procedures of the proposed STIGO framework are schematically illustrated in Figure 3. The inputs to the proposed STIGO framework include the computational model $\mathcal{M}$, the physical system which can be used for generating observations at any location, the prior density p_0 of model parameters, and the variance σ_{n}^2 of measurement noise ϵ. The parameters of this algorithm framework include the initial training data size N_0, the sample size M for MC

estimation, the Z-score β, the chain length L and scaling parameter η for generating posterior samples with R-MH algorithm, as well the four stopping thresholds ϵ_{STIGO}, ϵ_{BO}, ϵ_{BQ} and ϵ_{BMU}. The instructions for setting the parameters have been discussed as accompaniments to Algorithm 1 and Algorithm 2.

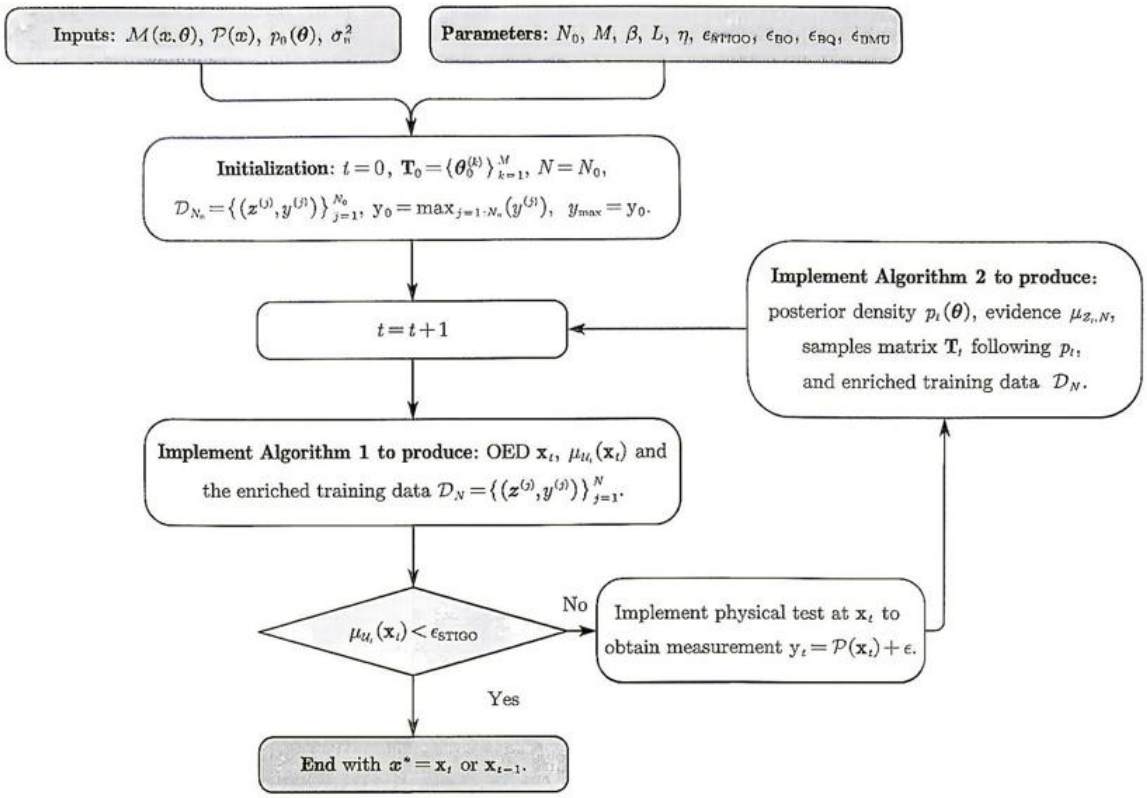


Figure 3: Flowchart of numerical implementation of the STIGO framework .

The stopping condition for the whole STIGO framework is formulated with Eq. (6), but practically is realized using the mean estimate of the utility function as:

$$\mu_{\mathcal{U}_t}(\mathbf{x}_t) < \epsilon_{\text{STIGO}} \tag{31}$$

, where the value of $\mu_{\mathcal{U}_t}(\mathbf{x}_t)$ is produced by Algorithm 1, and the value of ϵ_{STIGO} is dependent on σ_{n}. Practically, ϵ_{STIGO} is suggested to be $0.2 \sim 0.5\sigma_{\text{n}}$.

4. Benchmark Studies

This section presents three examples to empirically demonstrate STIGO. For each example, the true objective function $\mathcal{P}$ is provided solely for generating experimental measurements and estimating the ground

truth of global maximum. The algorithm parameters are set as the same for all examples: $N_0 = 15$, $M = 5 \times 10^3$, $\beta = 1$, $L = 30$, $\eta = 0.2$, $\epsilon_{\text{STIGO}} = 0.5\sigma_{\text{n}}$, $c_{\text{BO}} = 10^{-3}$, $\epsilon_{\text{BQ}} = 0.05$, and $\epsilon_{\text{BMU}} = 0.05$. For BOED, the CABO algorithm terminates when Eqs. (15) and (16) are satisfied simultaneously; while for BMU, the BQ algorithm terminates given that the stopping condition of Eq. (26) is satisfied consecutively for three times to avoid spurious convergence during the early training stage .

4.1. A two-dimensional optimization problem

We consider a numerical example modified from Ref. [25]. The computational model is formulated as:

$$\mathcal{M}(\boldsymbol{x}, \boldsymbol{\theta}) = \sum_{i=1}^{4} b_i \exp\left[-\theta_1 (x_1 - a_{i1})^2 - \theta_2 (x_2 - a_{i2})^2\right] \tag{32}$$

, where $a = \begin{pmatrix} -0.5 & 0.5 & -0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{pmatrix}^\top$ and $b = \begin{pmatrix} 1 & -1.5 & -1.5 & 2 \end{pmatrix}^\top$ are both known constants, $\boldsymbol{\theta} = (\theta_1, \theta_2)$ are two deterministic-but-unknown model parameters. The true objective function is formulated by setting model parameters as $\boldsymbol{\theta} = (\theta_1, \theta_2) = (2.5, 1.5)$, i.e., $\mathcal{P}(\boldsymbol{x}) = \mathcal{M}(\boldsymbol{x}, (2.5, 1.5))$. The feasible region of the design vector $\boldsymbol{x}$ is assumed to be $\mathbb{X} = [-1, 1] \times [-1, 1]$. Further, we assume that the physical test can be implemented at any location $\boldsymbol{x} \in \mathbb{X}$, by calling $\mathcal{P}(\boldsymbol{x})$ and adding a random error sampling from the Gaussian white noise with STD $\sigma_{\text{n}} = 0.005$. The prior distribution of $\boldsymbol{\theta}$ is assumed to be uniform within the support $\mathbb{T} = [0, 6] \times [0, 6]$. The target is to estimate the global maximum point $\boldsymbol{x}^* = \text{argmax}_{\boldsymbol{x} \in \mathbb{X}} \mathcal{P}(\boldsymbol{x})$. We use N to record the number of evaluating the model $\mathcal{M}$.

For one implementation of STIGO, four stages are produced, with the training details in the design variable space being reported in Figure 4. During each stage, one OED is produced by the CABO algorithm, and for the first three stages, the physical test is implemented at the resultant OEDs by calling $\mathcal{P}(\boldsymbol{x}) + \varepsilon$. The measurements are sequentially used for implementing BMU with the BQ algorithm, and the estimated prior/posterior densities for all stages are reported in Figure 5, together with the training data produced during each stage and the ground truth values of model parameters for comparison.

In Figure 4, the OED point produced for each stage and the corresponding utility function values are reported in the title of each subplot, and are compared in each subplot with the ground truth value of the global maximum point. The training samples produced during each stage are also reported in respective subplots. It is seen from Figure 4 that the true objective function $\mathcal{P}(\boldsymbol{x})$ includes two local minima and two local maxima, and at the forth stage, the CABO algorithm correctly produced the OED $\mathbf{x}_4 = (0.5469, 0.6316)$ with expected improvement $\mathcal{U}_4(\mathbf{x}_4) = 0.0004$ being less than the pre-specified threshold $\epsilon_{\text{STIGO}} = 0.0025$. The ground truth global maximum point of $\mathcal{P}(\boldsymbol{x})$ is (0.5509, 0.6290), indicating high accuracy of STIGO. As further shown, small part of the training samples are produced for global exploration, while the majority

truth of global maximum. The algorithm parameters are set as the same for all examples: $N_0 = 15$, $M = 5 \times 10^3$, $\beta = 1$, $L = 30$, $\eta = 0.2$, $\epsilon_{\text{STIGO}} = 0.5\sigma_{\text{n}}$, $\epsilon_{\text{BO}} = 10^{-3}$, $\epsilon_{\text{BQ}} = 0.05$, and $\epsilon_{\text{BMU}} = 0.05$. For BOED, the CABO algorithm terminates when Eqs. (15) and (16) are satisfied simultaneously; while for BMU, the BQ algorithm terminates given that the stopping condition of Eq. (26) is satisfied consecutively for three times to avoid spurious convergence during the early training stage .

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, where $\boldsymbol{a} = \begin{pmatrix} -0.5 & 0.5 & -0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{pmatrix}^{\mathsf{T}}$ and $b = \begin{pmatrix} 1 & -1.5 & -1.5 & 2 \end{pmatrix}^{\mathsf{T}}$ are both known constants, $\boldsymbol{\theta} = (\theta_1, \theta_2)$ are two deterministic-but-unknown model parameters. The true objective function is formulated by setting model parameters as $\boldsymbol{\theta} = (\theta_1, \theta_2) = (2.5, 1.5)$, i.e., $\mathcal{P}(\boldsymbol{x}) = \mathcal{M}(\boldsymbol{x}, (2.5, 1.5))$. The feasible region of the design vector $\boldsymbol{x}$ is assumed to be $\mathbb{X} = [-1, 1] \times [-1, 1]$. Further, we assume that the physical test can be implemented at any location $\boldsymbol{x} \in \mathbb{X}$, by calling $\mathcal{P}(\boldsymbol{x})$ and adding a random error sampling from the Gaussian white noise with STD $\sigma_{\text{n}} = 0.005$. The prior distribution of $\boldsymbol{\theta}$ is assumed to be uniform within the support $\mathbb{T} = [0, 6] \times [0, 6]$. The target is to estimate the global maximum point $\boldsymbol{x}^* = \text{argmax}_{\boldsymbol{x} \in \mathbb{X}} \mathcal{P}(\boldsymbol{x})$. We use N to record the number of evaluating the model $\mathcal{M}$.

For one implementation of STIGO, four stages are produced, with the training details in the design variable space being reported in Figure 4. During each stage, one OED is produced by the CABO algorithm, and for the first three stages, the physical test is implemented at the resultant OEDs by calling $\mathcal{P}(\boldsymbol{x}) + \varepsilon$. The measurements are sequentially used for implementing BMU with the BQ algorithm, and the estimated prior/posterior densities for all stages are reported in Figure 5, together with the training data produced during each stage and the ground truth values of model parameters for comparison.

In Figure 4, the OED point produced for each stage and the corresponding utility function values are reported in the title of each subplot, and are compared in each subplot with the ground truth value of the global maximum point. The training samples produced during each stage are also reported in respective subplots. It is seen from Figure 4 that the true objective function $\mathcal{P}(\boldsymbol{x})$ includes two local minima and two local maxima, and at the forth stage, the CABO algorithm correctly produced the OED $\mathbf{x}_4 = (0.5469, 0.6316)$ with expected improvement $\mathcal{U}_4(\mathbf{x}_4) = 0.0004$ being less than the pre-specified threshold $\epsilon_{\text{STIGO}} = 0.0025$. The ground truth global maximum point of $\mathcal{P}(\boldsymbol{x})$ is (0.5509, 0.6290), indicating high accuracy of STIGO. As further shown, small part of the training samples are produced for global exploration, while the majority

are distributed near the global optimum for exploitation. This indicates the high efficiency of CABO for addressing BOED.

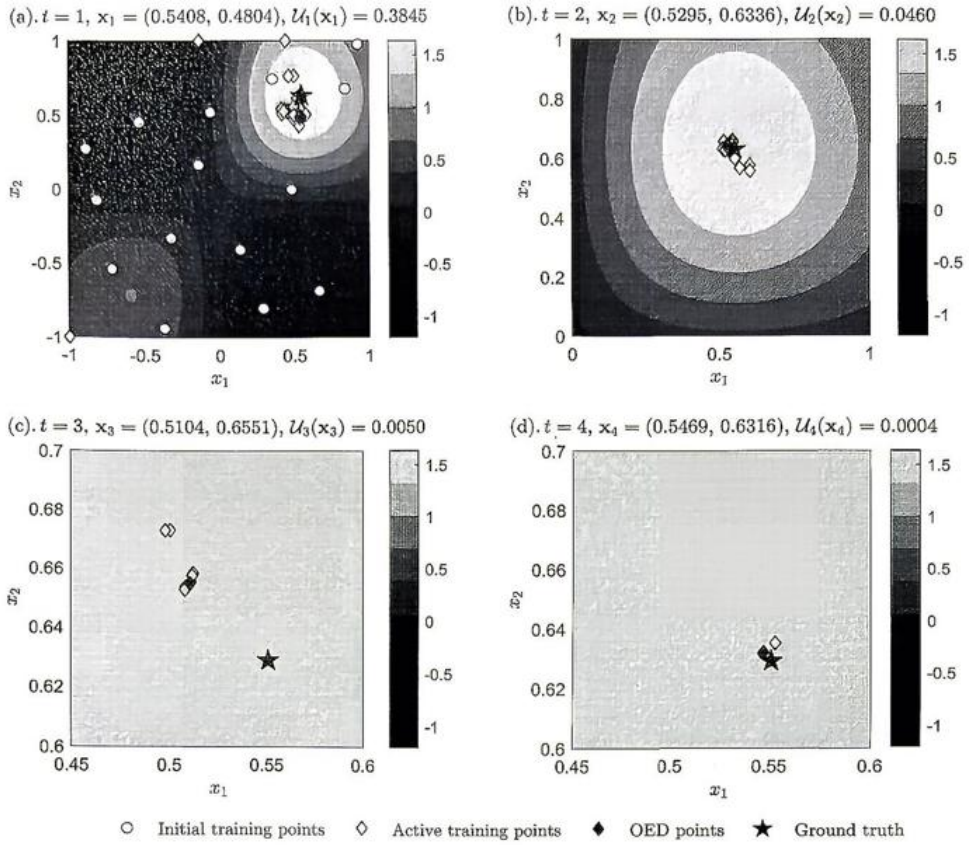


Figure 4: Training details in the design variable space for the 2D optimization problem. For each stage, the training points and OEDs are presented, together with the ground truth for comparison. The heatmaps visualize the true objective function $\mathcal{P}$. The generated OEDs and the corresponding utility function values are presented in the title of each subplot.

Figure 5(a) shows the prior density p_0, while the subsequent three subplots display the estimated posterior densities. It is seen that, with more measurements being fused, the posterior distribution show smaller variation, and the ground truth of the parameters are always located in the support of each posterior distribution, indicating high accuracy of BQ for numerically addressing BMU. It is also seen that most actively produced training data are located within the support of each posterior distribution, demonstrating the high effectiveness of the PVC acquisition function. Indeed, it is the high effectiveness of the PVC function enabling the proposed BQ algorithm to effectively tackle the BMU problem.

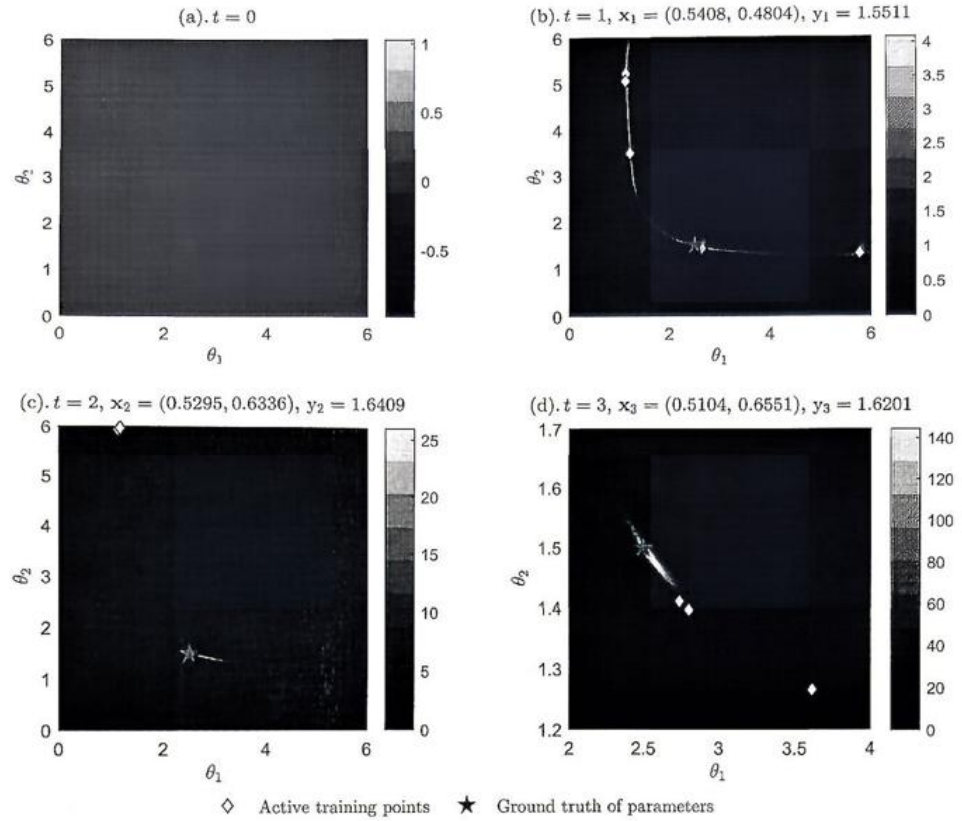


Figure 5: Heatmap plots for the mean estimates of the prior/posterior densities of the model parameters for the two-dimensional example, where the ground truth value of model parameters and the training points actively produced in each BMU procedure are also presented for illustration.

The effectiveness of the three acquisition functions enables the CABO and BQ algorithms numerically efficient for addressing the computational issues concerning BOED and BMU with as few training points as possible. Moreover, training data produced in each procedure are reused in the subsequent stages, rendering the STIGO framework highly efficient. Notably, only 94 model evaluations are consumed for generating the results in Figure 4 and Figure 5. To further verify the robustness of STIGO, the algorithm is implemented for five times, under the same parameter setting but with random initial training points, and the results are summarized in Table 1. It is seen that, among these five implementations, two of them consumed three physical tests, while the other three consumes four physical tests, to reach the same convergence criteria, but all produced accurate estimate of the global maxima. The highest number of consumed model calls is 109. These facts sufficiently demonstrate the robustness, accuracy and efficiency of STIGO.

Table 1: Summary of results for the 2D optimization problem, generated with five runs of STIGO.

Run No.	OEDs	Measurements	Max. $\mathcal{U}$-values	Ground truth	N
1	$x_1 = (0.5408, 0.4804)$	y_1 =1.5511	$\mathcal{U}_1 = 0.3845$		94
	$x_2 = (0.5295, 0.6336)$	y_2 =1.6409	$\mathcal{U}_2 = 0.0460$		
	$x_3 = (0.5104, 0.6551)$	y_3 =1.6201	$\mathcal{U}_3 = 0.0050$		
	$x_4 = (0.5469, 0.6316)$		$\mathcal{U}_4 = 0.0004$		
2	$x_1 = (0.4989, 0.5521)$	y_1 =1.5912	$\mathcal{U}_1 = 0.1682$		98
	$x_2 = (0.6314, 0.5044)$	y_2 =1.5490	$\mathcal{U}_2 = 0.0370$		
	$x_3 = (0.5579, 0.6220)$	y_3 =1.6368	$\mathcal{U}_3 = 0.0369$		
	$x_4 = (0.5524, 0.6263)$		$\mathcal{U}_4 = 0.0006$		
3	$x_1 = (0.5572, 0.5587)$	y_1 =1.6165	$\mathcal{U}_1 = 0.4734$	(0.5509, 0.6290)	101
	$x_2 = (0.5207, 0.6279)$	y_2 =1.6330	$\mathcal{U}_2 = 0.0232$		
	$x_3 = (0.5598, 0.6105)$	y_3 =1.6381	$\mathcal{U}_3 = 0.0097$		
	$x_4 = (0.5539, 0.6175)$	y_4 =1.6292	$\mathcal{U}_4 = 0.0035$		
	$x_5 = (0.5444, 0.6292)$		$\mathcal{U}_5 = 0.0017$		
4	$x_1 = (0.5135, 0.5054)$	y_1 =1.5718	$\mathcal{U}_1 = 0.4406$		109
	$x_2 = (0.5218, 0.6500)$	y_2 =1.6231	$\mathcal{U}_2 = 0.0403$		
	$x_3 = (0.5557, 0.6246)$	y_3 =1.6290	$\mathcal{U}_3 = 0.0096$		
	$x_4 = (0.5557, 0.6253)$	y_4 =1.6403	$\mathcal{U}_4 = 0.0038$		
	$x_5 = (0.5571, 0.6226)$		$\mathcal{U}_5 = 0.0003$		
5	$x_1 = (0.5303, 0.5218)$	y_1 =1.5888	$\mathcal{U}_1 = 0.1383$		97
	$x_2 = (0.5218, 0.6246)$	y_2 =1.6323	$\mathcal{U}_2 = 0.0329$		
	$x_3 = (0.5572, 0.6108)$	y_3 =1.6349	$\mathcal{U}_3 = 0.0066$		
	$x_4 = (0.5421, 0.6305)$	y_4 =1.6383	$\mathcal{U}_4 = 0.0039$		
	$x_5 = (0.5329, 0.6342)$		$\mathcal{U}_5 = 0.0021$		

4.2. A five-design-variable optimization problem

Consider a five-design-variable optimization problem with the model function formulated as:

$$\mathcal{M}(\boldsymbol{x}, \boldsymbol{\theta}) = \ln\left[\sum_{k=1}^{8} A_k \cdot \exp\left(-\frac{\| \boldsymbol{x} - \boldsymbol{c_k} \|^2}{2\sigma_{\boldsymbol{x}}^2}\right) \cdot \exp\left(-\frac{\| \boldsymbol{\theta} - \boldsymbol{d_k} \|^2}{2\sigma_{\boldsymbol{\theta}}^2}\right)\right] \tag{33}$$

, where $\boldsymbol{x}$ is the five-dimensional design vector, $\boldsymbol{\theta}$ indicates the three-dimensional deterministic-but-unknown model parameters, $\sigma_{\boldsymbol{x}} = 0.24$ and $\sigma_{\boldsymbol{\theta}} = 0.20$ are smoothing parameters, A_k, $\boldsymbol{c_k}$ and $\boldsymbol{d_k}$ $(k = 1, 2, \cdots, 8)$ are

all deterministic and known parameters with the values reported in Table 2. The true objective function is formulated as $\mathcal{P}(\boldsymbol{x}) = \mathcal{M}(\boldsymbol{x}, \boldsymbol{\theta}^*)$, where $\boldsymbol{\theta}^* = (0.5.\ 0.5,\ 0.5)$. With the above settings, this objective function exhibits eight local peaks, bring challenge for searching the global maximum. The prior distribution of each model parameter θ_i is assumed to be uniform with support [0, 1], and the feasible region of each design variable x_i is supposed to be [0, 1]. The STD σ_{n} of the measurement noise is set as 0.005 for creating artificial experimental measurements.

Table 2: Values of the deterministic and known parameters of the five-design-variable example.

k	A_k	$\mathbf{c}_k$	$\boldsymbol{d}_k$
1	0.84	(0.15, 0.20, 0.15, 0.25, 0.20)	(0.50, 0.50, 0.50)
2	0.78	(0.80, 0.15, 0.25, 0.20, 0.25)	(0.48, 0.52, 0.50)
3	0.77	(0.20, 0.80, 0.20, 0.15, 0.25)	(0.52, 0.48, 0.50)
4	0.80	(0.85, 0.20, 0.80, 0.20, 0.15)	(0.50, 0.50, 0.52)
5	0.76	(0.15, 0.20, 0.20, 0.80, 0.20)	(0.50, 0.48, 0.50)
6	1.10	(0.35, 0.30, 0.40, 0.35, 0.30)	(0.25, 0.28, 0.25)
7	1.06	(0.65, 0.60, 0.65, 0.60, 0.60)	(0.28, 0.25, 0.30)
8	1.38	(0.75, 0.25, 0.75, 0.25, 0.70)	(0.75, 0.72, 0.75)

All the algorithm parameters are set to be the same as the last example. In one run of STIGO, Four stages are automatically produced before reaching the stopping condition of Eq. (31), and thus three physical tests are implemented. The results of BOED during these four stages are reported in Table 3, together with the ground truth global maximum point and maximum value listed in the last row for comparison. Besides, the samples following the posterior densities of the first three stages are presented in Figure 6, together with the true values of model parameters. Table 3 indicates that, with the three measurements generated and used for calibrating the model parameters, the ODE point $\mathbf{x}_4$ generated in the fourth stage is in good agreement with the reference global maximum point, indicating the high accuracy of the proposed method. It is also shown in Figure 6 that the model parameter uncertainty has also been sufficiently reduced with these measurements, and the support of the posterior density at each stage contains the true values of the model parameters with high probability mass. This proves the effectiveness of the BQ and R-MH algorithm for addressing the BMU problem in each stage. This implementation of STIGO consumes three physical tests and 281 calls of the computational models, demonstrating high efficiency of STIGO.

Table 3: Summary of the BOED results generated for one implementation of STIGO for the five-design-variable optimization problem, together with the ground truths listed in the last row for comparison.

Stage No. t	OEDs $\mathbf{x}_t$	Measurements y_t	Max. U_t
1	(0.3112, 0.2854, 0.2975, 0.3077, 0.2897)	-0.2555	0.4960
2	(0.3534, 0.2972, 0.3838, 0.3719, 0.3018)	-0.5286	0.0609
3	(0.1810, 0.2452, 0.1771, 0.3222, 0.2212)	0.0025	0.1542
4	(0.1764, 0.2262, 0.1727, 0.2953, 0.2078)	-	0.0019
Ground truth	(0.1769, 0.2252, 0.1716, 0.2952, 0.2082)	0.0038	-

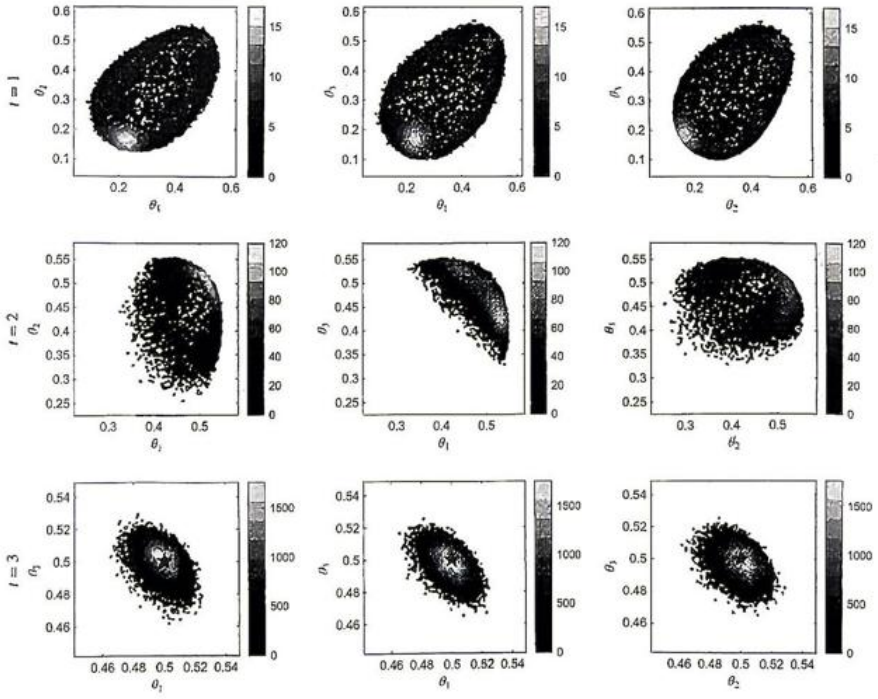


Figure 6: Posterior samples generated for the five-design-variable examples, with each row showing the pairwise samples for one stage, and the heatmaps indicating the densities of samples. The red pentagrams indicate the true values of model parameters.

Further, results generated from five independent runs of STIGO are summarized in Table 4. It is seen that each run produces accurate estimate of the global maximum point with three or four physical tests, and the number of model calls varies from 281 to 368, demonstrating the high robustness, accuracy and effectiveness of STIGO.

Table 4: Summary of results for the five-design-variable problem, generated with five independent runs.

Run No.	Resultant maximum points	No. of tests	No. of model calls
1	(0.1764, 0.2262, 0.1727, 0.2953, 0.2078)	3	281
2	(0.1763, 0.2265, 0.1709, 0.2936, 0.2071)	3	310
3	(0.1758, 0.2260, 0.1712, 0.2956, 0.2076)	3	365
4	(0.1773, 0.2256, 0.1724, 0.2951, 0.2074)	4	368
5	(0.1767, 0.2262, 0.1703, 0.2943, 0.2081)	3	296

4.3. PDE-constrained acoustic optimization

Consider a two-dimensional rectangular acoustic cavity $\Omega = [0,1] \times [0,1]$ (in meters) filled with air. The walls are assumed to be rigid with sound-hard boundaries. Inside the cavity, two coherent point sources emit harmonic sound waves of the same frequency with a fixed phase difference. The wave reflects from the walls and interfere. The goal is to maximize the sound intensity (squared pressure amplitude) at a fixed observation point $\boldsymbol{x}_{\text{obs}} = (0.48, 0.52)$ by adjusting the location of the first source, denoted as $\boldsymbol{s}_1$, given the second source being fixed at $\boldsymbol{s}_2 = (0.7, 0.3)$.

Denote by $\psi(\boldsymbol{x}) \in \mathbb{C}$ the complex acoustic pressure field. Fundamentally, the wave propagation is governed by the in-homogeneous Helmholtz equation with an artificial damping term, i.e.,

$$-\nabla^2\psi(\boldsymbol{x}) - (s^2 + i\theta_1\nu)\psi(\boldsymbol{x}) = \theta_2[\delta(\boldsymbol{x} - \boldsymbol{s}_1) + \delta(\boldsymbol{x} - \boldsymbol{s}_2)], \quad \boldsymbol{x} \in \Omega \tag{34}$$

, where ∇^2 refers to the Laplacian operator, ν is the wave number, which is assumed to be 18, θ_1 indicates the damping coefficient representing viscous absorption in the air, θ_2 refers to the source strength (assumed to be equal for both sources), and $\delta(\cdot)$ is the bivariate Dirac delta function. As all the four walls are assumed to be rigid, the boundary condition is represented by

$$-\frac{\partial\psi}{\partial\mathbf{n}} = 0, \quad \boldsymbol{x} \in \partial\Omega \tag{35}$$

, where $\mathbf{n}$ indicates the outward unit normal. The model is solved with Green's function modal superposition method, which expresses the solution as a double series of cosine eigenfunctions, each is truncated at the 80th orders to ensure accuracy.

With the above modeling, the coordinate values of the first source, denoted as x_1 and x_2 respectively, are set as design variables. The damping coefficient θ_1 and the source strength θ_2 are set as deterministic-but-unknown parameters with uniform prior distributions supported by $[0.2, 0.8]$ and $[0.9, 3.0]$ respectively. The true values of θ_1 and θ_2 are 0.25 and 1 respectively. The objective is to maximize the sound intensity at the observation point $\boldsymbol{x}_{\text{obs}} = (0.48, 0.52)$, i.e., $\mathcal{P}(x_1, x_2) = |\psi(\boldsymbol{x}_{\text{obs}}; x_1, x_2)|^2$, with the design variables

453 x_1 and x_2 both bounded by [0.1, 0.9]. The STD of measurement noise is assumed to be 0.005.

454 The true objective function is shown in Figure 7, together with the true global maximum point. It is seen
455 that five local maxima emerge, among which three sub-peaks' objective function values are very close to the
456 global peak, but each local maximum point is distant from the others. The problem is elaborately designed
457 in this way to test whether STIGO is capable of distinguishing similar modes and precisely identifying the
458 global maximum point. Furthermore, high initial uncertainty levels are imposed on both model parameters,
459 to examine the ability of STIGO against model uncertainties in searching for the global optimum.

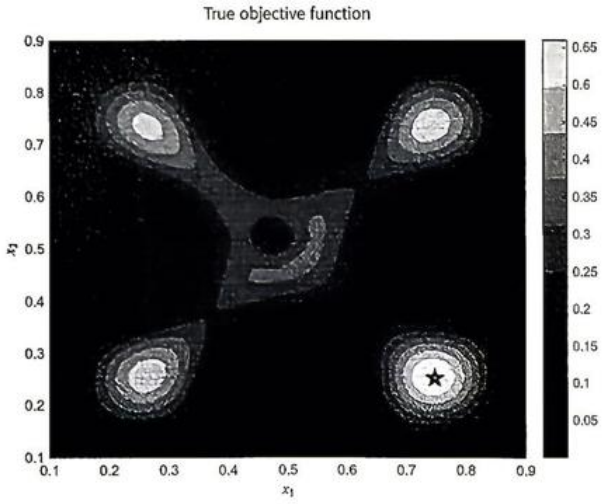


Figure 7: The true objective function of the PDE-constrained acoustic optimization problem.

460 All algorithm parameters are set as the same with the last two examples. The results generated by
461 one run of STIGO are summarized in Figure 8, with the first column shows the results of BMU, and the
462 second column displays the training details of BOED. Besides, results generated by five independent runs
463 of STIGO are reported in Table 5 for illustrating the robustness of the method.

464 Figure 8 shows that, during this implementation, the algorithm automatically produces three stages
465 before reaching the stopping condition, and thus two physical tests are implemented. The final posterior
466 density of model parameters (at stage $t = 2$) exhibits pronounced multimodality and approximate linear
467 correlation. Overall, the uncertainty of θ_2 is reduced more compared with that of θ_1, indicating that θ_1
468 is less sensitive than θ_2 for informing the global optimum. This constitutes one of the major distinctions
469 between the BOED embedded in STIGO and conventional BOED methods, which are designed to minimize
470 either model parameters or predictive uncertainties (see, e.g., [34, 21]). Specifically, BOED in STIGO aims
471 at searching the global optimum with sufficient credibility, and there is no requirement to fully suppress
472 the uncertainties associated with model parameters and/or predictions to an adequately low level. It is
473 also seen from Figure 8 (b) and (c), despite the considerable uncertainty presented in the two posterior

densities, the true values of the model parameters fall within the support domains of the posterior densities, demonstrating the numerical accuracy of proposed BQ algorithm for addressing the BMU task.

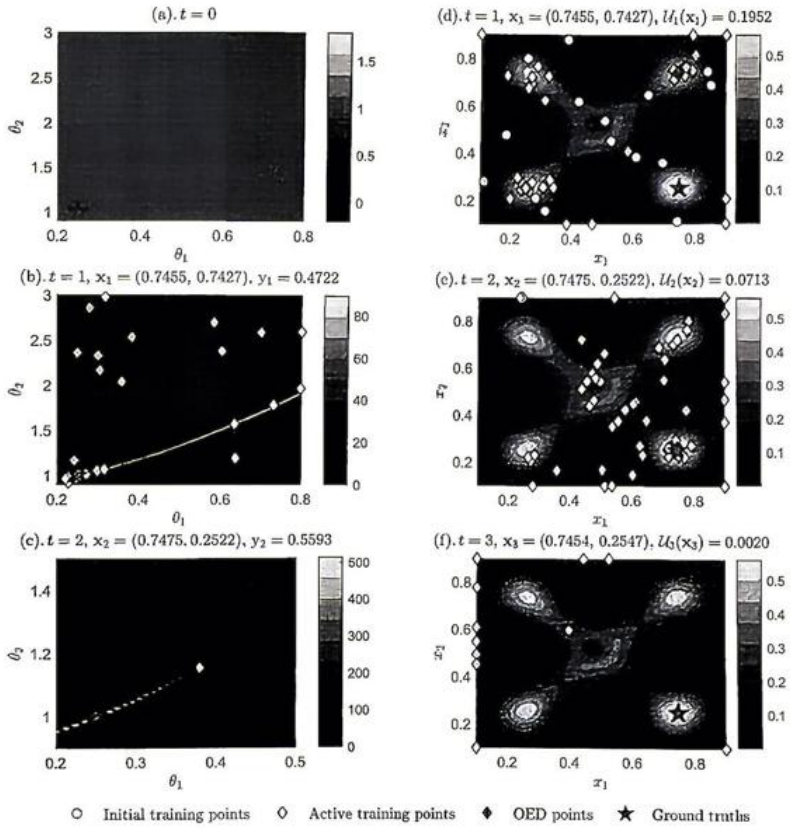


Figure 8: Visualization of the algorithm iteration process for the acoustic optimization problem, where (a)-(c) refer to the mean estimates of the prior/posterior densities, generated by BMU, accompanied with the training data and the true values of model parameters; (d)-(f) shows the true objective function with heatmaps, together with the training data and the OEDs produced by BOED, as well as the ground truth of the global maximum point.

From Figure 8(d), it is seen that, before receiving any experimental measurement, the CABO algorithm produce the ODE as a local maximum point, i.e., $\mathbf{x}_1$, owing to the extensive uncertainty of model parameters, as well as the fact that local peak values are very close to the global peak value. Then, from Figure 8(e), with the measurement $\mathrm{y}_1 = 0.4722$ observed at $\mathbf{x}_1$ being received and used for updating the model parameter uncertainty, the CABO procedure at the second stage successfully identifies the global maximum point to be $\mathbf{x}_2 = (0.7475, 0.2522)$. Further, with one more observation $\mathrm{y}_2 = 0.5593$ generated at $\mathbf{x}_2$, the model parameter is further calibrated, with which, the CABO algorithm at the third stage evaluate the maximum value of the utility function to be $\mathcal{U}_3(\mathbf{x}_3) = 0.0020$, which is smaller than the threshold $0.5\sigma_{\mathrm{n}}$, indicating that the global maximum point is identified with sufficient credibility. From Figure 8(f), it can be observed

that, for numerical implementation of BOED for the last stage (e.g., t=3), a number of training samples also emerge along the boundary of the feasible region of the design variables $\boldsymbol{x}$, although the true objective value at these boundaries are significantly smaller than the global peak values. This arises partly because multiple local peaks with objective values close to the global maximum exist in the vicinity of the boundary but outside of the feasible region.

Table 5: Results for the acoustic optimization problem, generated from five independent runs of STIGO.

Run No.	OEDs	Measurements	Max. $\mathcal{U}$-values	Ground truth	N
1	$\mathrm{x}_1 = (0.7455, 0.7427)$ $\mathrm{x}_2 = (0.7475, 0.2522)$ $\mathrm{x}_3 = (0.7454, 0.2547)$	y_1 =0.4722 y_2 =0.5593	$\mathcal{U}_1 = 0.1952$ $\mathcal{U}_2 = 0.0713$ $\mathcal{U}_3 = 0.0020$		183
2	$\mathrm{x}_1 = (0.5157, 0.4617)$ $\mathrm{x}_2 = (0.7365, 0.2491)$ $\mathrm{x}_3 = (0.7454, 0.2546)$ $\mathrm{x}_4 = (0.7455, 0.2545)$ $\mathrm{x}_5 = (0.7455, 0.2546)$	y_1 =0.3201 y_2 =0.5564 y_3 =0.5571 y_4=0.5512	$\mathcal{U}_1 = 0.1963$ $\mathcal{U}_2 = 0.1587$ $\mathcal{U}_3 = 0.0058$ $\mathcal{U}_4 = 0.0037$ $\mathcal{U}_5 = 0.0020$		140
3	$\mathrm{x}_1 = (0.5157, 0.4617)$ $\mathrm{x}_2 = (0.7470, 0.2523)$ $\mathrm{x}_3 = (0.7453, 0.2551)$ $\mathrm{x}_4 = (0.7454, 0.2551)$	y_1 =0.3201 y_2 =0.5589 y_3 =0.5596	$\mathcal{U}_1 = 0.1963$ $\mathcal{U}_2 = 0.1620$ $\mathcal{U}_3 = 0.0027$ $\mathcal{U}_4 = 0.0022$	(0.7426,0.2546)	136
4	$\mathrm{x}_1 = (0.5157, 0.4617)$ $\mathrm{x}_2 = (0.7459, 0.2518)$ $\mathrm{x}_3 = (0.7453, 0.2550)$	y_1 =0.3201 y_2 =0.5695	$\mathcal{U}_1 = 0.1963$ $\mathcal{U}_2 = 0.1617$ $\mathcal{U}_3 = 0.0020$		130
5	$\mathrm{x}_1 = (0.2645, 0.7406)$ $\mathrm{x}_2 = (0.7488, 0.2575)$ $\mathrm{x}_3 = (0.7460, 0.2540)$ $\mathrm{x}_4 = (0.7459, 0.2541)$	y_1 =0.4596 y_2 =0.5612 y_3 =0.5665	$\mathcal{U}_1 = 0.3233$ $\mathcal{U}_2 = 0.1388$ $\mathcal{U}_3 = 0.0038$ $\mathcal{U}_4 = 0.0016$		182

From the results reported in Table 5, it can be seen that all the five implementations successfully identify the global maximum point, with number of physical tests varying from 2 to 4, and the number model calls varying from 130 to 183. Compared with the results of the two-dimensional numerical example reported in Table 1, the number of required experimental runs in the present example is roughly equivalent, yet the number of model valuations is significantly higher. This is attributed to the three suboptimal peaks

in this example, which create strong deceptive effects. To navigate these deceptive regions and locate the the global maximum, STIGO expends extra model evaluations around these suboptimal peaks. Even so, the required number of experimental runs does not increase, leading to equivalent experimental cost, but higher computational cost. This sufficiently demonstrates the global convergence and high efficiency of STIGO, which is adapted from the theoretical global convergence of the traditional EI-based BO algorithm, and is numerically guaranteed by the effectiveness of the proposed STIGO framework as well as the specifically designed CABO and BQ algorithms.

5. Conclusions, Discussions and Prospects

To summary, STIGO has been developed for searching the global optima of a physical system by integrating the physical test with measurement noises and the computational model with parameter uncertainties. This framework is fundamentally based on implementing BOED and BMU alternately and sequentially, each realized with a specifically developed Bayesian numerical algorithm, for approaching the global optima with as few cost (in terms of the numbers of implementing physical tests and evaluating the computational model) as possible. To refine this theory and render it applicable to scenarios where simulation models are expensive to evaluate, a number of novel concepts and algorithms, including the utility function, the CABO algorithm for implementing BOED, the BQ and R-MH algorithm for implementing BMU, as well as the well-designed scheme for reusing training data, have been presented. The effectiveness of STIGO for searching the global optimum and updating the model parameters has been demonstrated with numerical and PDE-constrained optimization problems.

Compared with the traditional BO algorithms, of which the training data are labeled with either high-fidelity computational models (assumed to be precise, but rarely being true) or physical tests, the proposed STIGO framework is capable of fully exploiting experimental and computational resources to realize global optimization with minimal cost. Considering the remarkable success of the traditional BO algorithms across various domains, like synthetic chemistry and material discovery, the proposed STIGO framework is believed to possess great potential for practical applications therein.

The proposed STIGO framework includes multiple assumptions regarding physical tests and computational models, which, may not always hold in practical applications. Specifically, the measurement error is assumed to be Gaussian white noise with variance being exactly known, however, it can be practically any input-dependent stochastic process, which may not even be Gaussian and may not be known in advance (see e.g., [35]); while for computational models, it is assumed that the model uncertainty is governed by the unknownness of model parameters, however, unknown model bias may also exist in practices, as revealed by the classical Kennedy–O'Hagan framework [36]. Further generalization of STIGO is required to tackle complex application scenarios with those relaxed assumptions. Besides, the proposed STIGO framework

is limited to single-objective and unconstrained global optimization, and generalizations to multi-objective and/or constrained optimizations, where both objective and constraint functions can be labeled with physical tests and non-deterministic computational models, are also required to achieve wider applicability.

For numerical implementation, the developed CABO and BQ algorithms are both constructed based on GP model, which makes it unsuitable for high-dimensional tasks with, e.g., more than 20 design variables and model parameters. Although some advanced GP model, like additive GP and deep GP can be introduced for alleviating this limitation, their applicability are mostly problem-dependent. Basically, the numerical computational issues concerning STIGO, specifically the BOED and the BMU problems, are not inherently coupled with GPs. This is a big difference between STIGO and traditional BO. Both numerical tasks can be addressed with, e.g., sampling methods, which apply to high-dimensional cases, but with much more model calls. This gap on computational issue needs to be filled in future research.

Acknowledgment

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Appendix A. Closed-form expression of u-function

Based on the formulation of utility function with Eq. (7), the u-function is defined as:

$$u(\boldsymbol{z}) = \int_{\mathrm{R}} \max\left(\mathcal{M}(\boldsymbol{z}) - y_{\max} + \varepsilon, 0\right) \phi\left(\varepsilon | 0, \sigma_{\mathrm{n}}^2\right) \mathrm{d}\varepsilon \tag{A.1}$$

, where the non-negative integrand is non-zero if and only if $\varepsilon > y_{\max} - \mathcal{M}(\boldsymbol{z})$. Denote by $q = \mathcal{M}(\boldsymbol{z}) - y_{\max}$ and $w = \varepsilon/\sigma_{\mathrm{n}}$, the u-function can be reformulated as:

$$u(\boldsymbol{z}) = \int_{-q/\sigma_{\mathrm{n}}}^{\infty} (\sigma_{\mathrm{n}} w + q)\, \phi(w)\, \mathrm{d}w \tag{A.2}$$

, where the first and second parts can be respectively derived as:

$$\sigma_{\mathrm{n}} \int_{-q/\sigma_{\mathrm{n}}}^{\infty} w\phi(w)\, \mathrm{d}w = \sigma_{\mathrm{n}} \phi\left(-\frac{q}{\sigma_{\mathrm{n}}}\right) = \sigma_{\mathrm{n}} \phi\left(\frac{\mathcal{M}(\boldsymbol{z}) - y_{\max}}{\sigma_{\mathrm{n}}}\right) \tag{A.3a}$$

, and

$$q \int_{-q/\sigma_{\mathrm{n}}}^{\infty} \phi(w)\, \mathrm{d}w = q\Phi\left(\frac{q}{\sigma_{\mathrm{n}}}\right) = \left(\mathcal{M}(\boldsymbol{z}) - y_{\max}\right) \phi\left(\frac{\mathcal{M}(\boldsymbol{z}) - y_{\max}}{\sigma_{\mathrm{n}}}\right). \tag{A.3b}$$

Substituting Eq. (A.3) into Eq. (A.2) yields:

$$u(\boldsymbol{z}) = \left(\mathcal{M}(\boldsymbol{z}) - y_{\max}\right) \Phi\left(\frac{\mathcal{M}(\boldsymbol{z}) - y_{\max}}{\sigma_{\mathrm{n}}}\right) + \sigma_{\mathrm{n}} \phi\left(\frac{\mathcal{M}(\boldsymbol{z}) - y_{\max}}{\sigma_{\mathrm{n}}}\right). \tag{A.4}$$

□

Appendix B. Mathematical properties of u-function

For simplicity, denote $q = \left(\mathcal{M}(\boldsymbol{z}) - y_{\max}\right)/\sigma_{\mathrm{n}}$. The u-function against q can be formulated as:

$$u(q) = \sigma_{\mathrm{n}} q \Phi(q) + \sigma_{\mathrm{n}} \phi(q). \tag{B.1}$$

Further, the derivative of u over q is formulated as:

$$u'(q) = \sigma_{\mathrm{n}} \Phi(q) \geqslant 0 \tag{B.2}$$

, indicating that u is monotonically increasing w.r.t. q, and thus also $\mathcal{M}$. Besides, as shown by Eq. Eq. (A.1), the integrand for defining the u-function is non-negative, indicating that u is consistently non-negative.

□

Appendix C. Closed-form expressions for posterior moments of likelihood surrogate

Given the GP model $\mathcal{M}_N(\boldsymbol{z}) \sim \mathcal{GP}(\mu_{\mathcal{M}}(\boldsymbol{z}), c_{\mathcal{M}}(\boldsymbol{z}, \tilde{\boldsymbol{z}}))$ with $\boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{\theta})$, and a fixed OED $\mathbf{x}_t$, the likelihood of observation y_t is given by:

$$\mathcal{L}_N(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}) = \phi(\mathrm{y}_t - \mathcal{M}_N(\mathbf{x}_t, \boldsymbol{\theta})|0, \sigma_{\mathrm{n}}^2). \tag{C.1}$$

For a fixed $\mathbf{x}_t$, this likelihood forms a stochastic process indexed by $\boldsymbol{\theta}$, inheriting randomness from the GP posterior. For any given $\boldsymbol{\theta}$, the GP prediction at $(\mathbf{x}_t, \boldsymbol{\theta})$ follows a univariate Gaussian distribution, i.e., $\mathcal{M}_N(\mathbf{x}_t, \boldsymbol{\theta}) \sim \mathcal{N}(\mu_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}), \sigma^2_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}))$, where $\sigma^2_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}) = c_{\mathcal{M}}((\mathbf{x}_t, \boldsymbol{\theta}), (\mathbf{x}_t, \boldsymbol{\theta}))$ is the GP posterior variance.

The posterior mean prediction of the likelihood is defined as:

$$\mu_{\mathcal{L}}(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}) = \mathbb{E}_N[\phi(\mathrm{y}_t - \mathcal{M}_N|0, \sigma_{\mathrm{n}}^2)] \tag{C.2}$$

, where $\mathbb{E}_N[\cdot]$ represents the expectation operator w.r.t. the GP model. $\phi(\mathrm{y}_t - M|0, \sigma_{\mathrm{n}}^2)$ corresponds to the marginal density of $\mathrm{y}_t = \mathcal{M}_N + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_{\mathrm{n}}^2)$ is independent of $\mathcal{M}_N$. By the closure property of Gaussian distributions under summation, $\mathrm{y}_t \sim \mathcal{N}(\mu_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}), \sigma^2_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}) + \sigma_{\mathrm{n}}^2)$, and further using the convolution of two Gaussian distributions yields:

$$\begin{aligned} \mu_{\mathcal{L}}(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}) &= \int_{-\infty}^{\infty} \phi(\mathrm{y}_t - \mathcal{M}_N|\sigma_{\mathrm{n}}^2)\, \phi(\mathcal{M}_N|\mu_{\mathcal{M}}, \sigma^2_{\mathcal{M}})\, \mathrm{d}\mathcal{M}_N \\ &= \phi(\mathrm{y}_t|\mu_{\mathcal{M}}, \sigma^2_{\mathcal{M}} + \sigma_{\mathrm{n}}^2) \end{aligned}. \tag{C.3}$$

The prediction variance of the likelihood is formulated as:

$$\sigma^2_{\mathcal{L}}(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}) = \mathbb{E}_N[\mathcal{L}^2_N(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta})] - \mu^2_{\mathcal{L}}(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}) \tag{C.4}$$

, where $\mathcal{L}^2_N(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta})$ is expressed as:

$$\mathcal{L}^2_N(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}) = \phi^2(\mathrm{y}_t - \mathcal{M}_N|0, \sigma_{\mathrm{n}}^2) = \frac{1}{2\sqrt{\pi\sigma_{\mathrm{n}}^2}}\, \phi\left(\mathrm{y}_t - \mathcal{M}_N|0, \frac{\sigma_{\mathrm{n}}^2}{2}\right). \tag{C.5}$$

Following the same principle of Eq. (C.3), we can obtain:

$$\mathbb{E}_N[\mathcal{L}^2_N(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta})] = \frac{1}{2\sqrt{\pi\sigma_{\mathrm{n}}^2}}\, \phi\left(\mathrm{y}_t|\mu_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}),\ \sigma^2_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}) + \frac{\sigma_{\mathrm{n}}^2}{2}\right). \tag{C.6}$$

Substituting Eq. (C.6) into Eq. (C.4) yields:

$$\sigma^2_{\mathcal{L}}(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}) = \frac{1}{2\sqrt{\pi\sigma_{\mathrm{n}}^2}}\, \phi\left(\mathrm{y}_t|\mu_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}), \sigma^2_{\mathcal{M}}(\mathbf{x}_t, \boldsymbol{\theta}) + \frac{\sigma_{\mathrm{n}}^2}{2}\right) - \mu^2_{\mathcal{L}}(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta}). \tag{C.7}$$

For two distinct parameter values $\boldsymbol{\theta}$ and $\tilde{\boldsymbol{\theta}}$, $\mathcal{M}_N(\mathbf{x}_t, \boldsymbol{\theta})$ and $\mathcal{M}_N(\mathbf{x}_t, \tilde{\boldsymbol{\theta}})$ are jointly Gaussian with mean $\boldsymbol{\mu}_{\mathcal{M},\mathrm{biv}} =$

$[\mu_{\mathcal{M}}(\mathbf{x}_t,\theta),\mu_{\mathcal{M}}(\mathbf{x}_t,\tilde{\theta})]^\top$ and covariance

$$\mathbf{\Sigma}_{\mathcal{M},\text{biv}} = \begin{pmatrix} \sigma^2_{\mathcal{M}}(\mathbf{x}_t,\theta) & c_{\mathcal{M}}((\mathbf{x}_t,\theta),(\mathbf{x}_t,\tilde{\theta})) \\ c_{\mathcal{M}}((\mathbf{x}_t,\theta),(\mathbf{x}_t,\tilde{\theta})) & \sigma^2_{\mathcal{M}}(\mathbf{x}_t,\tilde{\theta}) \end{pmatrix}. \tag{C.8}$$

The posterior covariance $c_{\mathcal{L}}(\mathbf{x}_t,\mathrm{y}_t|\theta,\tilde{\theta})$ of the likelihood is then formulated as:

$$c_{\mathcal{L}}(\mathbf{x}_t,\mathrm{y}_t|\theta,\tilde{\theta}) = \mathbb{E}_N\left[\mathcal{L}_N(\mathbf{x}_t,\mathrm{y}_t|\theta)\mathcal{L}_N(\mathbf{x}_t,\mathrm{y}_t|\tilde{\theta})\right] - \mu_{\mathcal{L}}(\mathbf{x}_t,\mathrm{y}_t|\theta)\mu_{\mathcal{L}}(\mathbf{x}_t,\mathrm{y}_t|\tilde{\theta}). \tag{C.9}$$

Define $\mathcal{V}_N = \mathcal{M}_N(\mathbf{x}_t,\theta)+\varepsilon_1$ and $\tilde{\mathcal{V}}_N = \mathcal{M}_N\left(\mathbf{x}_t,\tilde{\theta}\right)+\varepsilon_2$, where $\varepsilon_1,\varepsilon_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\sigma_{\text{n}}^2)$. The mean and covariance of $\left(\mathcal{V}_N,\tilde{\mathcal{V}}_N\right)^\top$ are derived as $\boldsymbol{\mu}_{\mathcal{M},\text{biv}}$ and $\mathbf{\Sigma}_{\mathcal{M},\text{biv}}+\sigma_{\text{n}}^2\mathbf{I}_2$, with I_2 indicating the second-order identity matrix. Denoting by $\phi_2(\cdot|\boldsymbol{\mu}_{\text{M,biv}},\mathbf{\Sigma}_{\text{M}},\text{biv}+\sigma_{\text{n}}^2\mathbf{I}_2)$ a bivariate Gaussian density, the expectation $\mathbb{E}_N\left[\mathcal{L}_N(\mathbf{x}_t,\mathrm{y}_t|\theta)\mathcal{L}_N(\mathbf{x}_t,\mathrm{y}_t|\tilde{\theta})\right]$ can be formulated, using the convolution of two bivariate Gaussian distributions, as:

$$\mathbb{E}_N\left[\mathcal{L}_N(\mathbf{x}_t,\mathrm{y}_t|\theta)\mathcal{L}_N(\mathbf{x}_t,\mathrm{y}_t|\tilde{\theta})\right] = \phi_2\left(\mathrm{y}_t\mathbf{1}_{2\times1}\,|\,\boldsymbol{\mu}_{\mathcal{M},\text{biv}},\,\mathbf{\Sigma}_{\mathcal{M},\text{biv}}+\sigma_{\text{n}}^2\mathbf{I}_2\right) \tag{C.10}$$

, where $\mathbf{1}_{2\times1}$ implies the 2-by-1 matrix of ones. Substituting Eq. (C.10) into Eq. (C.9) we can obtain the closed-form expression of $c_{\mathcal{L}}\left(\mathbf{x}_t,\mathrm{y}_t|\theta,\tilde{\theta}\right)$.

□

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